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# Fresh vs Rotten Fruit Classification - Multi-Task Learning
# Complete Conference-Ready Implementation with Enhanced Preprocessing
and Improved ViT
# FINAL VERSION WITH ALL ISSUES FIXED FOR 9 MODELS
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from tqdm import tqdm
import os
import warnings
warnings.filterwarnings('ignore')

# Deep Learning Libraries
import tensorflow as tf
from tensorflow import keras
from tensorflow.keras import layers, models, optimizers, regularizers
from tensorflow.keras.applications import VGG16, InceptionV3,
ResNet152V2
from tensorflow.keras.applications import EfficientNetB4, DenseNet121
from tensorflow.keras.applications import ConvNeXtBase
from tensorflow.keras.preprocessing.image import ImageDataGenerator
from tensorflow.keras.callbacks import EarlyStopping, ModelCheckpoint,
ReduceLROnPlateau
from sklearn.utils.class_weight import compute_class_weight

# FIXED: Add tensorflow_hub import for ViT model
import tensorflow_hub as hub

# Metrics and Evaluation
from sklearn.metrics import classification_report, confusion_matrix,
roc_curve, auc
from sklearn.preprocessing import LabelEncoder
from sklearn.model_selection import train_test_split

# Set style for publication-quality figures
plt.style.use('default')
sns.set_palette("husl")
plt.rcParams.update({
    'font.size': 12,
    'axes.titlesize': 14,
    'axes.labelsize': 12,
    'xtick.labelsize': 10,
    'ytick.labelsize': 10,
    'legend.fontsize': 10,
    'figure.titlesize': 16
})

# Set random seeds for reproducibility
np.random.seed(42)

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tf.random.set_seed(42)

# GPU configuration
gpus = tf.config.experimental.list_physical_devices('GPU')
if gpus:
    try:
        for gpu in gpus:
            tf.config.experimental.set_memory_growth(gpu, True)
        print(f"GPU available: {len(gpus)} GPU(s)")
    except RuntimeError as e:
        print(e)
else:
    print("GPU not available, using CPU")

print(f"TensorFlow version: {tf.__version__}")
print(f"Keras version: {keras.__version__}")

#=====
# ENHANCED DATA LOADING AND PREPROCESSING - FIXED VERSION
#=====

def normalize_fruit_name(folder_name):
    """
    Enhanced fruit name normalization with comprehensive spelling
    fixes
    """
    folder_lower = folder_name.lower().strip()

    # Remove fresh/rotten prefixes
    if folder_lower.startswith('fresh'):
        fruit_name = folder_lower[5:].strip()
        freshness = 1 # Fresh = 1
    elif folder_lower.startswith('rotten'):
        fruit_name = folder_lower[6:].strip()
        freshness = 0 # Rotten = 0
    else:
        fruit_name = folder_lower
        # Try to infer freshness from name
        if 'rotten' in fruit_name or 'bad' in fruit_name:
            freshness = 0
        else:
            freshness = 1 # Default to fresh

    # Comprehensive spelling corrections and normalization
    corrections = {
        'tamto': 'tomato',
        'patato': 'potato',
        'bittergroud': 'bittergourd',

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        'apsicum': 'capsicum',
        'apples': 'apple',
        'oranges': 'orange',
        'bananas': 'banana',
        'tomatos': 'tomato',
        'potatos': 'potato',
        'cucumbers': 'cucumber'
    }

    fruit_name = corrections.get(fruit_name, fruit_name)

    return fruit_name, freshness

def create_dummy_data():
    """Create dummy data for testing when actual dataset is not
    available"""
    print("△ Creating dummy dataset for testing...")
    fruits = ['apple', 'banana', 'orange', 'tomato', 'potato',
              'capsicum', 'cucumber', 'okra', 'bittergourd']
    dummy_data = []

    for fruit in fruits:
        for fresh in [0, 1]:
            for i in range(200): # Increased from 150 to 200
                dummy_data.append({
                    'filepath': f'dummy_{fruit}_{fresh}_{i}.jpg',
                    'fruit': fruit,
                    'fresh': fresh
                })

    df = pd.DataFrame(dummy_data)
    print(f"□ Dummy dataset created with {len(df)} samples")
    return df

def load_and_process_dataset(base_path='/kaggle/input/fresh-and-stale-
classification'):
    """
    Enhanced dataset loading with comprehensive preprocessing pipeline
    - FIXED VERSION
    """
    print("□ Starting Enhanced Dataset Loading and Processing...")

    TRAIN_PATH = os.path.join(base_path, 'dataset/Train')
    TEST_PATH = os.path.join(base_path, 'dataset/Test')

    def load_data_from_path(path):
        """Load data from a specific path"""
        filepaths, fruits, freshness = [], [], []

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        if not os.path.exists(path):
            print(f"△ Path {path} does not exist")
            return pd.DataFrame({'filepath': [], 'fruit': [], 'fresh':
[]})

        try:
            folders = [f for f in os.listdir(path) if
os.path.isdir(os.path.join(path, f))]
        except Exception as e:
            print(f"△ Error reading directory {path}: {e}")
            return pd.DataFrame({'filepath': [], 'fruit': [], 'fresh':
[]})

        for folder in tqdm(folders, desc=f"Processing
{os.path.basename(path)}"):
            try:
                folder_path = os.path.join(path, folder)
                fruit_name, fresh_value = normalize_fruit_name(folder)

                # Get all image files
                image_files = [f for f in os.listdir(folder_path)
                               if f.lower().endswith(('.png', '.jpg',
'.jpeg', '.bmp', '.tiff'))]

                for img_file in image_files:
                    full_path = os.path.join(folder_path, img_file)
                    # Only add if file actually exists
                    if os.path.exists(full_path):
                        filepaths.append(full_path)
                        fruits.append(fruit_name)
                        freshness.append(fresh_value)
            except Exception as e:
                print(f"△ Error processing folder {folder}: {e}")
                continue

        return pd.DataFrame({
            'filepath': filepaths,
            'fruit': fruits,
            'fresh': freshness
        })

# Load train and test data
df_train = load_data_from_path(TRAIN_PATH)
df_test = load_data_from_path(TEST_PATH)

# Combine datasets
df_all = pd.concat([df_train, df_test], axis=0, ignore_index=True)

if len(df_all) == 0:

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        print("⚠ No data found, creating dummy data...")
        return create_dummy_data()

    print(f"📁 Total images loaded: {len(df_all)}")
    print(f"📁 Fruit distribution:")
    fruit_counts = df_all['fruit'].value_counts()
    print(fruit_counts)

    # Target 9 fruit classes
    target_fruits = ['apple', 'banana', 'capsicum', 'cucumber',
                    'okra', 'orange', 'potato', 'tomato', 'bittergourd']

    # Filter to target fruits only
    df_filtered = df_all[df_all['fruit'].isin(target_fruits)].copy()

    if len(df_filtered) == 0:
        print("⚠ No target fruits found, creating dummy data...")
        return create_dummy_data()

    print(f"\n📁 Filtered to target fruits: {len(df_filtered)} images")
    print("📁 Target fruit distribution:")
    filtered_counts = df_filtered['fruit'].value_counts()
    print(filtered_counts)

    return df_filtered

def apply_smart_balancing(df):
    """
    Apply smart balancing strategy - FIXED VERSION
    """
    print("\n📁 Applying Smart Balancing Strategy...")

    if len(df) == 0:
        print("⚠ Empty dataframe received, creating dummy data...")
        return create_dummy_data()

    balanced_data = []
    augmentation_stats = {}

    for fruit in df['fruit'].unique():
        fruit_data = df[df['fruit'] == fruit].copy()
        total_samples = len(fruit_data)

        print(f"\n📁 Processing {fruit}: {total_samples} samples")

        if total_samples >= 800: # Reduced threshold from 1000 to 800
            # Case 1: Sufficient data, sample up to 3000
            target_size = min(3000, total_samples)

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# Stratified sampling to maintain fresh/rotten balance
fresh_data = fruit_data[fruit_data['fresh'] == 1]
rotten_data = fruit_data[fruit_data['fresh'] == 0]

if len(fresh_data) == 0:
    fresh_ratio = 0
elif len(rotten_data) == 0:
    fresh_ratio = 1
else:
    fresh_ratio = len(fresh_data) / total_samples

target_fresh = max(1, int(target_size * fresh_ratio))
target_rotten = target_size - target_fresh

sampled_fresh = fresh_data.sample(n=min(target_fresh,
len(fresh_data)),
                                random_state=42,
replace=len(fresh_data) < target_fresh)
sampled_rotten = rotten_data.sample(n=min(target_rotten,
len(rotten_data)),
                                   random_state=42,
replace=len(rotten_data) < target_rotten)

fruit_balanced = pd.concat([sampled_fresh,
sampled_rotten])
print(f"   📦 Sampled {len(fruit_balanced)} images (Fresh:
{len(sampled_fresh)}, Rotten: {len(sampled_rotten)})")

else:
    # Case 2: Insufficient data, augment to reach minimum
threshold
samples
    target_size = max(800, total_samples) # Minimum 800
samples

    if total_samples < 100:
        print(f"   ⚠️ WARNING: {fruit} has only {total_samples}
original images (<100)")

    # Calculate how many augmented images we need
needed_samples = target_size - total_samples

    if needed_samples > 0:
        print(f"   📦 Need to augment {needed_samples}
additional images")

    # Create augmented entries (not actual images, just
metadata)
    augmented_data = []
    samples_per_original = max(1, needed_samples //

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total_samples)

    for _, row in fruit_data.iterrows():
        for i in range(samples_per_original):
            aug_row = row.copy()
            # Mark as augmented with unique identifier
            aug_row['filepath'] =
f"{row['filepath']}_aug_{i}"
            aug_row['is_augmented'] = True
            augmented_data.append(aug_row)

        # Add remaining samples if needed
        remaining = needed_samples - len(augmented_data)
        if remaining > 0:
            for i in range(remaining):
                orig_idx = i % len(fruit_data)
                aug_row = fruit_data.iloc[orig_idx].copy()
                aug_row['filepath'] =
f"{aug_row['filepath']}_aug_extra_{i}"
                aug_row['is_augmented'] = True
                augmented_data.append(aug_row)

            # Mark original data
            fruit_data_marked = fruit_data.copy()
            fruit_data_marked['is_augmented'] = False

            # Combine original and augmented
            fruit_balanced = pd.concat([fruit_data_marked,
pd.DataFrame(augmented_data)]).reset_index(drop=True)
            print(f"    Created {len(fruit_balanced)} samples
({total_samples} original + {len(augmented_data)} augmented)")

            augmentation_stats[fruit] = {
                'original': total_samples,
                'augmented': len(augmented_data),
                'total': len(fruit_balanced)
            }
        else:
            fruit_data['is_augmented'] = False
            fruit_balanced = fruit_data
            print(f"    Using {len(fruit_balanced)} original
samples")

        balanced_data.append(fruit_balanced)

    # Combine all balanced data
    df_balanced = pd.concat(balanced_data, ignore_index=True)

    # Ensure is_augmented column exists

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if 'is_augmented' not in df_balanced.columns:
    df_balanced['is_augmented'] = False

print(f"\n Final Balanced Dataset Statistics:")
print(f"Total samples: {len(df_balanced)}")
print("\nFruit distribution:")
print(df_balanced['fruit'].value_counts())
print("\nFreshness distribution:")
print(df_balanced['fresh'].value_counts())

if augmentation_stats:
    print(f"\n Augmentation Statistics:")
    for fruit, stats in augmentation_stats.items():
        print(f" {fruit}: {stats['original']} → {stats['total']}
(+{stats['augmented']} augmented)")

    return df_balanced

def create_stratified_splits(df):
    """
    Create 70/15/15 stratified splits - FIXED VERSION
    """
    print("\n Creating Stratified Train/Val/Test Splits
(70/15/15)...")

    if len(df) == 0:
        print("△ Empty dataframe for splitting")
        return pd.DataFrame(), pd.DataFrame(), pd.DataFrame()

    # Create stratification key
    df['stratify_key'] = df['fruit'] + '_' + df['fresh'].astype(str)

    # Check if we have enough samples for stratification
    stratify_counts = df['stratify_key'].value_counts()
    min_count = stratify_counts.min()

    if min_count < 3: # Need at least 3 samples for 70/15/15 split
        print(f"△ Some classes have too few samples ({min_count}).
Using random splits...")
        train_df, temp_df = train_test_split(df, test_size=0.3,
random_state=42)
        val_df, test_df = train_test_split(temp_df, test_size=0.5,
random_state=42)
    else:
        try:
            # First split: 70% train, 30% temp
            train_df, temp_df = train_test_split(
                df, test_size=0.3, stratify=df['stratify_key'],
random_state=42

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    )

    # Second split: 15% val, 15% test from the 30% temp
    val_df, test_df = train_test_split(
        temp_df, test_size=0.5,
        stratify=temp_df['stratify_key'], random_state=42
    )

    except ValueError as e:
        print(f"⚠ Stratification failed: {e}. Using random
splits...")
        train_df, temp_df = train_test_split(df, test_size=0.3,
        random_state=42)
        val_df, test_df = train_test_split(temp_df, test_size=0.5,
        random_state=42)

    # Remove stratify_key column
    for split_df in [train_df, val_df, test_df]:
        if 'stratify_key' in split_df.columns:
            split_df.drop('stratify_key', axis=1, inplace=True)

    print(f"📊 Train: {len(train_df)} ({len(train_df)/len(df)*100:.1f}
%)")
    print(f"📊 Val: {len(val_df)} ({len(val_df)/len(df)*100:.1f}%)")
    print(f"📊 Test: {len(test_df)} ({len(test_df)/len(df)*100:.1f}%)")

    return train_df, val_df, test_df

def compute_class_weights(train_df):
    """
    Compute class weights for both fruit and freshness tasks - FIXED
    VERSION
    """
    print("\n🔢 Computing Class Weights...")

    if len(train_df) == 0:
        print("⚠ Empty training dataframe, using default weights")
        return {}, {0: 1.0, 1: 1.0}

    try:
        # Fruit class weights
        fruit_classes = np.unique(train_df['fruit'])
        if len(fruit_classes) > 1:
            fruit_weights = compute_class_weight(
                'balanced',
                classes=fruit_classes,
                y=train_df['fruit']
            )
            fruit_class_weights = dict(zip(range(len(fruit_classes)),

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fruit_weights))
    else:
        fruit_class_weights = {0: 1.0}

        # Freshness class weights
        fresh_classes = np.unique(train_df['fresh'])
        if len(fresh_classes) > 1:
            fresh_weights = compute_class_weight(
                'balanced',
                classes=fresh_classes,
                y=train_df['fresh']
            )
            fresh_class_weights = dict(zip(fresh_classes,
fresh_weights))
        else:
            fresh_class_weights = {0: 1.0, 1: 1.0}

        print(f" Fruit class weights: {fruit_class_weights}")
        print(f" Fresh class weights: {fresh_class_weights}")

        return fruit_class_weights, fresh_class_weights

except Exception as e:
    print(f"△ Error computing class weights: {e}")
    return {}, {0: 1.0, 1: 1.0}

#=====
#
# ENHANCED DATA GENERATORS - FIXED VERSION
#=====
#

class EnhancedMultiTaskDataGenerator(keras.utils.Sequence):
    """Enhanced data generator with augmentation support - FIXED
    VERSION"""

    def __init__(self, dataframe, datagen, target_size=(224, 224),
batch_size=32,
                fruit_classes=None, shuffle=True):
        self.dataframe = dataframe.reset_index(drop=True)
        self.datagen = datagen
        self.target_size = target_size
        self.batch_size = batch_size
        self.fruit_classes = fruit_classes if fruit_classes is not
None else []
        self.num_classes = len(self.fruit_classes)
        self.shuffle = shuffle
        self.indices = np.arange(len(self.dataframe))

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        # Ensure we have valid data
        if len(self.dataframe) == 0:
            print("⚠ Warning: Empty dataframe provided to data
generator")

        self.on_epoch_end()

    def __len__(self):
        if len(self.dataframe) == 0:
            return 1 # Return at least 1 to avoid division by zero
        return int(np.ceil(len(self.dataframe) / self.batch_size))

    def __getitem__(self, index):
        # Handle empty dataframe
        if len(self.dataframe) == 0:
            return self._get_dummy_batch()

        # Get batch indices
        start_idx = index * self.batch_size
        end_idx = min((index + 1) * self.batch_size,
len(self.dataframe))
        batch_indices = self.indices[start_idx:end_idx]

        # Handle case where batch_indices is empty
        if len(batch_indices) == 0:
            return self._get_dummy_batch()

        batch_df = self.dataframe.iloc[batch_indices]

        batch_images = []
        batch_fruit_labels = []
        batch_fresh_labels = []

        for _, row in batch_df.iterrows():
            try:
                # Handle both real and augmented images
                is_augmented = row.get('is_augmented', False)

                if is_augmented and '_aug_' in str(row['filepath']):
                    # Extract original filepath for augmented images
                    original_path =
str(row['filepath']).split('_aug_')[0]
                    img_array =
self._load_and_augment_image(original_path)
                else:
                    img_array = self._load_image(row['filepath'])

                if img_array is not None:
                    batch_images.append(img_array)

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        # Get fruit label index
        if row['fruit'] in self.fruit_classes:
            fruit_idx =
self.fruit_classes.index(row['fruit'])
        else:
            fruit_idx = 0 # Default to first class if not
found

            batch_fruit_labels.append(fruit_idx)
            batch_fresh_labels.append(int(row['fresh']))

    except Exception as e:
        print(f"△ Error processing row: {e}")
        # Add dummy data for failed samples
        dummy_img = self._get_dummy_image()
        batch_images.append(dummy_img)
        batch_fruit_labels.append(0)
        batch_fresh_labels.append(0)

    # Ensure we have at least one sample
    if len(batch_images) == 0:
        return self._get_dummy_batch()

    # Convert to arrays
    batch_images = np.array(batch_images)
    batch_fruit_labels = np.array(batch_fruit_labels)
    batch_fresh_labels = np.array(batch_fresh_labels)

    # Convert fruit labels to categorical
    if self.num_classes > 0:
        batch_fruit_labels = tf.keras.utils.to_categorical(
            batch_fruit_labels, num_classes=self.num_classes
        )
    else:
        # Handle case with no fruit classes
        batch_fruit_labels = np.zeros((len(batch_images), 1))

    return batch_images, {
        'fruit_output': batch_fruit_labels,
        'fresh_output': batch_fresh_labels
    }

def _get_dummy_batch(self):
    """Generate a dummy batch when no data is available"""
    dummy_images = np.random.rand(1, *self.target_size, 3) * 255
    if self.num_classes > 0:
        dummy_fruit = np.zeros((1, self.num_classes))
        dummy_fruit[0, 0] = 1 # Set first class to 1
    else:

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        dummy_fruit = np.array([[0.0]])
        dummy_fresh = np.array([0.0])

        return dummy_images, {
            'fruit_output': dummy_fruit,
            'fresh_output': dummy_fresh
        }

    def _get_dummy_image(self):
        """Generate a dummy image"""
        return np.random.rand(*self.target_size, 3) * 255

    def _load_image(self, filepath):
        """Load and preprocess a single image - FIXED VERSION"""
        try:
            if os.path.exists(filepath):
                img = tf.keras.preprocessing.image.load_img(
                    filepath, target_size=self.target_size
                )
                img_array =
tf.keras.preprocessing.image.img_to_array(img)
                img_array = self.datagen.standardize(img_array)
                return img_array
            else:
                # Return dummy image if file doesn't exist
                return self._get_dummy_image()
        except Exception as e:
            print(f"⚠ Error loading {filepath}: {e}")
            return self._get_dummy_image()

    def _load_and_augment_image(self, filepath):
        """Load image and apply augmentation - FIXED VERSION"""
        try:
            if os.path.exists(filepath):
                img = tf.keras.preprocessing.image.load_img(
                    filepath, target_size=self.target_size
                )
                img_array =
tf.keras.preprocessing.image.img_to_array(img)

                # Apply random augmentation
                img_array = img_array.reshape((1,) + img_array.shape)

                # Create a flow and get one augmented image
                aug_iter = self.datagen.flow(img_array, batch_size=1)
                try:
                    aug_img = next(aug_iter)[0] # FIXED: Use next()
instead of .next()
                except (StopIteration, IndexError):

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        # Fallback to original image if augmentation fails
        aug_img = img_array[0]

        return self.datagen.standardize(aug_img)
    else:
        return self._get_dummy_image()
except Exception as e:
    print(f"△ Error augmenting {filepath}: {e}")
    return self._get_dummy_image()

def on_epoch_end(self):
    if self.shuffle:
        np.random.shuffle(self.indices)

#=====
# ENHANCED MODEL DEFINITIONS - FIXED VERSION
#=====

def create_multi_task_custom_cnn(input_shape=(224, 224, 3),
num_fruit_classes=9):
    """Custom CNN for Multi-Task Learning - FIXED VERSION"""

    # Ensure minimum number of classes
    num_fruit_classes = max(1, num_fruit_classes)

    inputs = layers.Input(shape=input_shape, name='input_layer')

    # Shared backbone with proper error handling
    x = layers.Conv2D(32, (3, 3), activation='relu', padding='same',
name='conv1_1')(inputs)
    x = layers.BatchNormalization(name='bn1_1')(x)
    x = layers.Conv2D(32, (3, 3), activation='relu', padding='same',
name='conv1_2')(x)
    x = layers.MaxPooling2D((2, 2), name='pool1')(x)
    x = layers.Dropout(0.25, name='dropout1')(x)

    x = layers.Conv2D(64, (3, 3), activation='relu', padding='same',
name='conv2_1')(x)
    x = layers.BatchNormalization(name='bn2_1')(x)
    x = layers.Conv2D(64, (3, 3), activation='relu', padding='same',
name='conv2_2')(x)
    x = layers.MaxPooling2D((2, 2), name='pool2')(x)
    x = layers.Dropout(0.25, name='dropout2')(x)

    x = layers.Conv2D(128, (3, 3), activation='relu', padding='same',
name='conv3_1')(x)
    x = layers.BatchNormalization(name='bn3_1')(x)

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    x = layers.Conv2D(128, (3, 3), activation='relu', padding='same',
name='conv3_2')(x)
    x = layers.MaxPooling2D((2, 2), name='pool3')(x)
    x = layers.Dropout(0.25, name='dropout3')(x)

    x = layers.Conv2D(256, (3, 3), activation='relu', padding='same',
name='conv4_1')(x)
    x = layers.BatchNormalization(name='bn4_1')(x)
    x = layers.Conv2D(256, (3, 3), activation='relu', padding='same',
name='conv4_2')(x)
    x = layers.MaxPooling2D((2, 2), name='pool4')(x)
    x = layers.Dropout(0.25, name='dropout4')(x)

    # Shared dense layers
    x = layers.Flatten(name='flatten')(x)
    x = layers.Dense(512, activation='relu', name='dense1')(x)
    x = layers.BatchNormalization(name='bn_dense1')(x)
    x = layers.Dropout(0.5, name='dropout_dense1')(x)
    shared_features = layers.Dense(256, activation='relu',
name='shared_features')(x)

    # Fruit classification head
    fruit_branch = layers.Dense(128, activation='relu',
name='fruit_dense')(shared_features)
    fruit_branch = layers.Dropout(0.3, name='fruit_dropout')
(fruit_branch)
    fruit_output = layers.Dense(num_fruit_classes,
activation='softmax', name='fruit_output')(fruit_branch)

    # Freshness classification head
    fresh_branch = layers.Dense(64, activation='relu',
name='fresh_dense')(shared_features)
    fresh_branch = layers.Dropout(0.3, name='fresh_dropout')
(fresh_branch)
    fresh_output = layers.Dense(1, activation='sigmoid',
name='fresh_output')(fresh_branch)

    model = models.Model(inputs=inputs, outputs=[fruit_output,
fresh_output], name='Custom_CNN_MultiTask')
    return model

def create_multi_task_vgg16(input_shape=(224, 224, 3),
num_fruit_classes=9):
    """VGG16 for Multi-Task Learning - FIXED VERSION"""
    num_fruit_classes = max(1, num_fruit_classes)

    try:
        base_model = VGG16(weights='imagenet', include_top=False,
input_shape=input_shape)

```

```

# Freeze early layers
for layer in base_model.layers[:-4]:
    layer.trainable = False

inputs = base_model.input
x = base_model.output
x = layers.GlobalAveragePooling2D(name='global_avg_pool')(x)
x = layers.Dense(512, activation='relu', name='dense_512')(x)
x = layers.BatchNormalization(name='bn_512')(x)
x = layers.Dropout(0.5, name='dropout_512')(x)
shared_features = layers.Dense(256, activation='relu',
name='shared_256')(x)

# Task-specific heads
fruit_branch = layers.Dense(128, activation='relu',
name='fruit_128')(shared_features)
fruit_branch = layers.Dropout(0.3, name='fruit_dropout_vgg')(
(fruit_branch))
fruit_output = layers.Dense(num_fruit_classes,
activation='softmax', name='fruit_output')(fruit_branch)

fresh_branch = layers.Dense(64, activation='relu',
name='fresh_64')(shared_features)
fresh_branch = layers.Dropout(0.3, name='fresh_dropout_vgg')(
(fresh_branch))
fresh_output = layers.Dense(1, activation='sigmoid',
name='fresh_output')(fresh_branch)

model = models.Model(inputs=inputs, outputs=[fruit_output,
fresh_output], name='VGG16_MultiTask')
return model
except Exception as e:
    print(f"Error creating VGG16 model: {e}")
    return create_multi_task_custom_cnn(input_shape,
num_fruit_classes)

def create_multi_task_inceptionv3(input_shape=(224, 224, 3),
num_fruit_classes=9):
    """InceptionV3 for Multi-Task Learning - FIXED VERSION"""
    num_fruit_classes = max(1, num_fruit_classes)

    try:
        base_model = InceptionV3(weights='imagenet',
include_top=False, input_shape=input_shape)

        for layer in base_model.layers[:-20]:
            layer.trainable = False

```

```

        inputs = base_model.input
        x = base_model.output
        x = layers.GlobalAveragePooling2D(name='global_avg_pool_inc')
(x)
        x = layers.Dense(512, activation='relu', name='dense_512_inc')
(x)
        x = layers.BatchNormalization(name='bn_512_inc')(x)
        x = layers.Dropout(0.5, name='dropout_512_inc')(x)
        shared_features = layers.Dense(256, activation='relu',
name='shared_256_inc')(x)

        fruit_branch = layers.Dense(128, activation='relu',
name='fruit_128_inc')(shared_features)
        fruit_branch = layers.Dropout(0.3, name='fruit_dropout_inc')
(fruit_branch)
        fruit_output = layers.Dense(num_fruit_classes,
activation='softmax', name='fruit_output')(fruit_branch)

        fresh_branch = layers.Dense(64, activation='relu',
name='fresh_64_inc')(shared_features)
        fresh_branch = layers.Dropout(0.3, name='fresh_dropout_inc')
(fresh_branch)
        fresh_output = layers.Dense(1, activation='sigmoid',
name='fresh_output')(fresh_branch)

        model = models.Model(inputs=inputs, outputs=[fruit_output,
fresh_output], name='InceptionV3_MultiTask')
        return model
    except Exception as e:
        print(f"Error creating InceptionV3 model: {e}")
        return create_multi_task_custom_cnn(input_shape,
num_fruit_classes)

def create_multi_task_resnet152(input_shape=(224, 224, 3),
num_fruit_classes=9):
    """ResNet152V2 for Multi-Task Learning - FIXED VERSION"""
    num_fruit_classes = max(1, num_fruit_classes)

    try:
        base_model = ResNet152V2(weights='imagenet',
include_top=False, input_shape=input_shape)

        for layer in base_model.layers[:-30]:
            layer.trainable = False

        inputs = base_model.input
        x = base_model.output
        x = layers.GlobalAveragePooling2D(name='global_avg_pool_res')
(x)

```

```

        x = layers.Dense(512, activation='relu', name='dense_512_res')
(x)
        x = layers.BatchNormalization(name='bn_512_res')(x)
        x = layers.Dropout(0.5, name='dropout_512_res')(x)
        shared_features = layers.Dense(256, activation='relu',
name='shared_256_res')(x)

        fruit_branch = layers.Dense(128, activation='relu',
name='fruit_128_res')(shared_features)
        fruit_branch = layers.Dropout(0.3, name='fruit_dropout_res')
(fruit_branch)
        fruit_output = layers.Dense(num_fruit_classes,
activation='softmax', name='fruit_output')(fruit_branch)

        fresh_branch = layers.Dense(64, activation='relu',
name='fresh_64_res')(shared_features)
        fresh_branch = layers.Dropout(0.3, name='fresh_dropout_res')
(fresh_branch)
        fresh_output = layers.Dense(1, activation='sigmoid',
name='fresh_output')(fresh_branch)

        model = models.Model(inputs=inputs, outputs=[fruit_output,
fresh_output], name='ResNet152_MultiTask')
        return model
    except Exception as e:
        print(f"Error creating ResNet152 model: {e}")
        return create_multi_task_custom_cnn(input_shape,
num_fruit_classes)

def create_darknet_conv_block(x, filters, kernel_size=3, strides=1):
    """Darknet convolutional block - FIXED VERSION"""
    x = layers.Conv2D(filters, kernel_size, strides=strides,
padding='same', use_bias=False)(x)
    x = layers.BatchNormalization()(x)
    x = layers.LeakyReLU(alpha=0.1)(x)
    return x

def create_darknet_residual_block(x, filters):
    """Darknet residual block - FIXED VERSION"""
    shortcut = x
    x = create_darknet_conv_block(x, filters//2, 1)
    x = create_darknet_conv_block(x, filters, 3)
    x = layers.Add()([shortcut, x])
    return x

def create_multi_task_yolov3_backbone(input_shape=(224, 224, 3),
num_fruit_classes=9):
    """YOLOv3 Darknet-53 backbone for Multi-Task Learning - FIXED
VERSION"""
    num_fruit_classes = max(1, num_fruit_classes)

```

```

inputs = layers.Input(shape=input_shape, name='darknet_input')

# Darknet-53 backbone (simplified)
x = create_darknet_conv_block(inputs, 32, 3)
x = create_darknet_conv_block(x, 64, 3, strides=2)
x = create_darknet_residual_block(x, 64)

x = create_darknet_conv_block(x, 128, 3, strides=2)
for _ in range(2):
    x = create_darknet_residual_block(x, 128)

x = create_darknet_conv_block(x, 256, 3, strides=2)
for _ in range(4):
    x = create_darknet_residual_block(x, 256)

x = create_darknet_conv_block(x, 512, 3, strides=2)
for _ in range(4):
    x = create_darknet_residual_block(x, 512)

# Shared features
x = layers.GlobalAveragePooling2D(name='darknet_global_pool')(x)
x = layers.Dense(128, activation='relu',
kernel_regularizer=regularizers.l2(0.01), name='darknet_dense_128')(x)
x = layers.Dropout(0.5, name='darknet_dropout_128')(x)
shared_features = layers.Dense(64, activation='relu',
kernel_regularizer=regularizers.l2(0.01), name='darknet_shared_64')(x)

# Task-specific heads
fruit_output = layers.Dense(num_fruit_classes,
activation='softmax', name='fruit_output')(shared_features)
fresh_output = layers.Dense(1, activation='sigmoid',
name='fresh_output')(shared_features)

model = models.Model(inputs=inputs, outputs=[fruit_output,
fresh_output], name='YOLOv3_MultiTask')
return model

def create_multi_task_efficientnet(input_shape=(224, 224, 3),
num_fruit_classes=9):
    """EfficientNetB4 for Multi-Task Learning"""
    from tensorflow.keras.applications import EfficientNetB4

    num_fruit_classes = max(1, num_fruit_classes)

    try:
        base_model = EfficientNetB4(weights='imagenet',
include_top=False, input_shape=input_shape)

```

```

    # Fine-tune the top layers
    for layer in base_model.layers[:-20]:
        layer.trainable = False

    inputs = base_model.input
    x = base_model.output
    x = layers.GlobalAveragePooling2D(name='global_avg_pool_eff')(x)
    x = layers.Dense(512, activation='relu', name='dense_512_eff')(x)
    x = layers.BatchNormalization(name='bn_512_eff')(x)
    x = layers.Dropout(0.3, name='dropout_512_eff')(x)
    shared_features = layers.Dense(256, activation='relu',
name='shared_256_eff')(x)

    # Task-specific heads
    fruit_branch = layers.Dense(128, activation='relu',
name='fruit_128_eff')(shared_features)
    fruit_branch = layers.Dropout(0.2, name='fruit_dropout_eff')(fruit_branch)
    fruit_output = layers.Dense(num_fruit_classes,
activation='softmax', name='fruit_output')(fruit_branch)

    fresh_branch = layers.Dense(64, activation='relu',
name='fresh_64_eff')(shared_features)
    fresh_branch = layers.Dropout(0.2, name='fresh_dropout_eff')(fresh_branch)
    fresh_output = layers.Dense(1, activation='sigmoid',
name='fresh_output')(fresh_branch)

    model = models.Model(inputs=inputs, outputs=[fruit_output,
fresh_output], name='EfficientNetB4_MultiTask')
    return model
except Exception as e:
    print(f"Error creating EfficientNet model: {e}")
    return create_multi_task_custom_cnn(input_shape,
num_fruit_classes)

def create_multi_task_densenet(input_shape=(224, 224, 3),
num_fruit_classes=9):
    """DenseNet121 for Multi-Task Learning"""
    from tensorflow.keras.applications import DenseNet121

    num_fruit_classes = max(1, num_fruit_classes)

    try:
        base_model = DenseNet121(weights='imagenet',
include_top=False, input_shape=input_shape)

```

```

    # Fine-tune later layers
    for layer in base_model.layers[:-15]:
        layer.trainable = False

    inputs = base_model.input
    x = base_model.output
    x =
layers.GlobalAveragePooling2D(name='global_avg_pool_dense')(x)
    x = layers.Dense(512, activation='relu',
name='dense_512_dense')(x)
    x = layers.BatchNormalization(name='bn_512_dense')(x)
    x = layers.Dropout(0.4, name='dropout_512_dense')(x)
    shared_features = layers.Dense(256, activation='relu',
name='shared_256_dense')(x)

    # Task-specific heads
    fruit_branch = layers.Dense(128, activation='relu',
name='fruit_128_dense')(shared_features)
    fruit_branch = layers.Dropout(0.3, name='fruit_dropout_dense')
(fruit_branch)
    fruit_output = layers.Dense(num_fruit_classes,
activation='softmax', name='fruit_output')(fruit_branch)

    fresh_branch = layers.Dense(64, activation='relu',
name='fresh_64_dense')(shared_features)
    fresh_branch = layers.Dropout(0.3, name='fresh_dropout_dense')
(fresh_branch)
    fresh_output = layers.Dense(1, activation='sigmoid',
name='fresh_output')(fresh_branch)

    model = models.Model(inputs=inputs, outputs=[fruit_output,
fresh_output], name='DenseNet121_MultiTask')
    return model
except Exception as e:
    print(f"Error creating DenseNet model: {e}")
    return create_multi_task_custom_cnn(input_shape,
num_fruit_classes)

def create_multi_task_convnext_base(input_shape=(224, 224, 3),
num_fruit_classes=9):
    """ConvNeXt-Base Pretrained for Multi-Task Learning"""
    from tensorflow.keras.applications import ConvNeXtBase

    num_fruit_classes = max(1, num_fruit_classes)

    try:
        base_model = ConvNeXtBase(weights='imagenet',
include_top=False, input_shape=input_shape)

```

```

        for layer in base_model.layers[:-10]:
            layer.trainable = False

        inputs = base_model.input
        x = base_model.output
        x = layers.GlobalAveragePooling2D(name='convnext_gap')(x)
        x = layers.Dense(512, activation='relu',
name='convnext_dense_512')(x)
        x = layers.BatchNormalization(name='convnext_bn_512')(x)
        x = layers.Dropout(0.3, name='convnext_dropout_512')(x)
        shared_features = layers.Dense(256, activation='relu',
name='convnext_shared_256')(x)

        # Task-specific heads
        fruit_branch = layers.Dense(128, activation='relu',
name='convnext_fruit_128')(shared_features)
        fruit_branch = layers.Dropout(0.2,
name='convnext_fruit_dropout')(fruit_branch)
        fruit_output = layers.Dense(num_fruit_classes,
activation='softmax', name='fruit_output')(fruit_branch)

        fresh_branch = layers.Dense(64, activation='relu',
name='convnext_fresh_64')(shared_features)
        fresh_branch = layers.Dropout(0.2,
name='convnext_fresh_dropout')(fresh_branch)
        fresh_output = layers.Dense(1, activation='sigmoid',
name='fresh_output')(fresh_branch)

        model = models.Model(inputs=inputs, outputs=[fruit_output,
fresh_output], name='ConvNeXtBase_MultiTask')
        return model
    except Exception as e:
        print(f"Error creating ConvNeXt model: {e}")
        return create_multi_task_custom_cnn(input_shape,
num_fruit_classes)

def create_multi_task_vit_base_pretrained(input_shape=(224, 224, 3),
num_fruit_classes=9):
    """Google's Pretrained ViT-Base-16 for Multi-Task Learning - Best
ViT Choice"""

    num_fruit_classes = max(1, num_fruit_classes)

    try:
        # Load pretrained ViT-Base-16 from TensorFlow Hub
        vit_base_url =
"https://tfhub.dev/google/vit\_base\_patch16\_224/1"
        base_model = hub.KerasLayer(vit_base_url, trainable=True)

        inputs = layers.Input(shape=input_shape, name='vit_input')

```

```

        # Pretrained ViT feature extraction (this has ImageNet-21k
        pretraining!)
        x = base_model(inputs) # Shape: (batch_size, 768)

        # Add custom multi-task heads optimized for your
        fruit/freshness tasks
        x = layers.Dense(512, activation='relu', name='vit_dense_512')(x)
        x = layers.BatchNormalization(name='vit_bn_512')(x)
        x = layers.Dropout(0.3, name='vit_dropout_512')(x)
        shared_features = layers.Dense(256, activation='relu',
        name='vit_shared_256')(x)

        # Task-specific heads
        fruit_branch = layers.Dense(128, activation='relu',
        name='vit_fruit_128')(shared_features)
        fruit_branch = layers.Dropout(0.2, name='vit_fruit_dropout')(
        fruit_branch)
        fruit_output = layers.Dense(num_fruit_classes,
        activation='softmax', name='fruit_output')(fruit_branch)

        fresh_branch = layers.Dense(64, activation='relu',
        name='vit_fresh_64')(shared_features)
        fresh_branch = layers.Dropout(0.2, name='vit_fresh_dropout')(
        fresh_branch)
        fresh_output = layers.Dense(1, activation='sigmoid',
        name='fresh_output')(fresh_branch)

        model = models.Model(inputs=inputs, outputs=[fruit_output,
        fresh_output], name='ViT_Base_Pretrained_MultiTask')
        return model
    except Exception as e:
        print(f"Error creating pretrained ViT model: {e}")
        return create_multi_task_custom_cnn(input_shape,
        num_fruit_classes)

#=====
# TRAINING AND EVALUATION FUNCTIONS - FIXED VERSION
#=====

def cleanup_memory():
    """Clean up memory between model training"""
    import gc
    try:
        gc.collect()
        tf.keras.backend.clear_session()

```

```

except:
    pass

def compile_and_train_multi_task_model(model, model_name, train_gen,
val_gen, epochs=15, alpha=0.7, class_weights=None):
    """Compile and train a multi-task model with class weights - FIXED
VERSION"""
    print(f"\n{'='*50}")
    print(f"Training {model_name}")
    print(f"{'='*50}")

    # Check if generators have data
    if len(train_gen) == 0 or len(val_gen) == 0:
        print("⚠ Warning: Empty data generators")
        # Create dummy history
        dummy_history = type('DummyHistory', (), {})(())
        dummy_history.history = {
            'loss': [1.0] * epochs,
            'val_loss': [1.0] * epochs,
            'fruit_output_accuracy': [0.1] * epochs,
            'val_fruit_output_accuracy': [0.1] * epochs,
            'fresh_output_accuracy': [0.5] * epochs,
            'val_fresh_output_accuracy': [0.5] * epochs,
            'fruit_output_loss': [1.0] * epochs,
            'val_fruit_output_loss': [1.0] * epochs,
            'fresh_output_loss': [0.5] * epochs,
            'val_fresh_output_loss': [0.5] * epochs
        }
        return model, dummy_history

    # Enhanced learning rate for different models
    if 'ViT' in model_name or 'Vision' in model_name:
        learning_rate = 0.0005
    elif 'ResNet' in model_name:
        learning_rate = 0.0005
    else:
        learning_rate = 0.001

    try:
        model.compile(
            optimizer=optimizers.Adam(learning_rate=learning_rate),
            loss={
                'fruit_output': 'categorical_crossentropy',
                'fresh_output': 'binary_crossentropy'
            },
            loss_weights={
                'fruit_output': alpha,
                'fresh_output': 1 - alpha
            },

```

```

        metrics={
            'fruit_output': ['accuracy'],
            'fresh_output': ['accuracy']
        }
    )

    callbacks = [
        EarlyStopping(
            monitor='val_loss',
            patience=5, # Reduced patience for faster training
            restore_best_weights=True,
            verbose=1
        ),
        ReduceLROnPlateau(
            monitor='val_loss',
            factor=0.5,
            patience=3,
            min_lr=1e-7,
            verbose=1
        )
    ]

    # Train the model with proper error handling
    history = model.fit(
        train_gen,
        epochs=epochs,
        validation_data=val_gen,
        callbacks=callbacks,
        verbose=1,
    )

    return model, history

except Exception as e:
    print(f"Error during training {model_name}: {e}")
    # Create dummy history
    dummy_history = type('DummyHistory', (), {}]()
    dummy_history.history = {
        'loss': [1.0] * epochs,
        'val_loss': [1.0] * epochs,
        'fruit_output_accuracy': [0.1] * epochs,
        'val_fruit_output_accuracy': [0.1] * epochs,
        'fresh_output_accuracy': [0.5] * epochs,
        'val_fresh_output_accuracy': [0.5] * epochs,
        'fruit_output_loss': [1.0] * epochs,
        'val_fruit_output_loss': [1.0] * epochs,
        'fresh_output_loss': [0.5] * epochs,
        'val_fresh_output_loss': [0.5] * epochs
    }

```

```

        return model, dummy_history

def evaluate_multi_task_model(model, train_gen, val_gen, test_gen,
model_name):
    """Evaluate multi-task model on all three sets - FIXED VERSION"""
    print(f"Evaluating {model_name}...")

    def safe_evaluate(generator, split_name):
        """Safely evaluate model on a generator"""
        try:
            if len(generator) == 0:
                print(f"△ Empty {split_name} generator")
                return {'loss': 999.0, 'fruit_output_accuracy': 0.0,
'fresh_output_accuracy': 0.0}

            results = model.evaluate(generator, verbose=0,
return_dict=True)
            return results
        except Exception as e:
            print(f"△ Error evaluating {split_name}: {e}")
            return {'loss': 999.0, 'fruit_output_accuracy': 0.0,
'fresh_output_accuracy': 0.0}

    train_results = safe_evaluate(train_gen, 'train')
    val_results = safe_evaluate(val_gen, 'validation')
    test_results = safe_evaluate(test_gen, 'test')

    results = {
        'Model': model_name,
        'Train Loss': train_results.get('loss', 999.0),
        'Valid Loss': val_results.get('loss', 999.0),
        'Test Loss': test_results.get('loss', 999.0),
        'Train Fruit Acc': train_results.get('fruit_output_accuracy',
0.0),
        'Valid Fruit Acc': val_results.get('fruit_output_accuracy',
0.0),
        'Test Fruit Acc': test_results.get('fruit_output_accuracy',
0.0),
        'Train Fresh Acc': train_results.get('fresh_output_accuracy',
0.0),
        'Valid Fresh Acc': val_results.get('fresh_output_accuracy',
0.0),
        'Test Fresh Acc': test_results.get('fresh_output_accuracy',
0.0)
    }

    return results

#=====

```

```

=====
# ALL VISUALIZATION FUNCTIONS - FIXED FOR 9 MODELS
#=====
=====

def save_individual_fruit_accuracy_comparison(results_df,
output_dir="conference_figures"):
    """Save individual fruit classification accuracy comparison"""
    if not os.path.exists(output_dir):
        os.makedirs(output_dir)

    plt.figure(figsize=(12, 8))

    models = results_df['Model']
    x_pos = np.arange(len(models))

    # FIXED: Updated colors for 9 models
    colors = ['#FF6B6B', '#4ECDC4', '#45B7D1', '#96CEB4', '#FECA57',
    '#FF9FF3', '#A8E6CF', '#FFD93D', '#FF8A80']
    bars = plt.bar(x_pos, results_df['Test Fruit Acc'],
color=colors[:len(models)], alpha=0.8, edgecolor='black', linewidth=1)

    for i, bar in enumerate(bars):
        height = bar.get_height()
        plt.text(bar.get_x() + bar.get_width()/2., height + 0.01,
            f'{height:.3f}', ha='center', va='bottom',
fontweight='bold', fontsize=11)

    plt.title('Fruit Classification Test Accuracy Comparison',
fontsize=16, fontweight='bold', pad=20)
    plt.xlabel('Deep Learning Models', fontsize=14, fontweight='bold')
    plt.ylabel('Test Accuracy', fontsize=14, fontweight='bold')
    plt.xticks(x_pos, models, rotation=45, ha='right', fontsize=12)
    plt.ylim(0, 1.1)
    plt.grid(True, alpha=0.3, axis='y')
    plt.tight_layout()

    filename = os.path.join(output_dir,
'fruit_classification_accuracy_comparison.png')
    plt.savefig(filename, dpi=300, bbox_inches='tight',
facecolor='white')
    plt.show()
    print(f"✓ Fruit accuracy comparison saved: {filename}")

def save_individual_freshness_accuracy_comparison(results_df,
output_dir="conference_figures"):
    """Save individual freshness classification accuracy comparison"""
    if not os.path.exists(output_dir):
        os.makedirs(output_dir)

```

```

plt.figure(figsize=(12, 8))

models = results_df['Model']
x_pos = np.arange(len(models))

# FIXED: Updated colors for 9 models
colors = ['#FF6B6B', '#4ECDC4', '#45B7D1', '#96CEB4', '#FECA57',
          '#FF9FF3', '#A8E6CF', '#FFD93D', '#FF8A80']
bars = plt.bar(x_pos, results_df['Test Fresh Acc'],
               color=colors[:len(models)], alpha=0.8, edgecolor='black', linewidth=1)

for i, bar in enumerate(bars):
    height = bar.get_height()
    plt.text(bar.get_x() + bar.get_width()/2., height + 0.01,
             f'{height:.3f}', ha='center', va='bottom',
             fontweight='bold', fontsize=11)

plt.title('Freshness Classification Test Accuracy Comparison',
         fontsize=16, fontweight='bold', pad=20)
plt.xlabel('Deep Learning Models', fontsize=14, fontweight='bold')
plt.ylabel('Test Accuracy', fontsize=14, fontweight='bold')
plt.xticks(x_pos, models, rotation=45, ha='right', fontsize=12)
plt.ylim(0, 1.1)
plt.grid(True, alpha=0.3, axis='y')
plt.tight_layout()

filename = os.path.join(output_dir,
                        'freshness_classification_accuracy_comparison.png')
plt.savefig(filename, dpi=300, bbox_inches='tight',
           facecolor='white')
plt.show()
print(f"✓ Freshness accuracy comparison saved: {filename}")

def save_individual_combined_performance_plot(results_df,
output_dir="conference_figures"):
    """Save individual combined performance comparison"""
    if not os.path.exists(output_dir):
        os.makedirs(output_dir)

    plt.figure(figsize=(12, 8))

    x = range(len(results_df))
    plt.plot(x, results_df['Test Fruit Acc'], 'o-', label='Fruit
Classification',
            linewidth=3, markersize=10, color='#FF6B6B')
    plt.plot(x, results_df['Test Fresh Acc'], 's-', label='Freshness
Classification',
            linewidth=3, markersize=10, color='#4ECDC4')

```

```

plt.title('Multi-Task Test Performance Comparison', fontsize=16,
fontweight='bold', pad=20)
plt.xlabel('Deep Learning Models', fontsize=14, fontweight='bold')
plt.ylabel('Test Accuracy', fontsize=14, fontweight='bold')
plt.xticks(x, results_df['Model'], rotation=45, ha='right',
fontsize=12)
plt.legend(fontsize=12, loc='best')
plt.grid(True, alpha=0.3)
plt.ylim(0, 1.1)
plt.tight_layout()

filename = os.path.join(output_dir,
'combined_performance_comparison.png')
plt.savefig(filename, dpi=300, bbox_inches='tight',
facecolor='white')
plt.show()
print(f"✓ Combined performance comparison saved: {filename}")

def save_individual_loss_comparison(results_df,
output_dir="conference_figures"):
    """Save individual loss comparison"""
    if not os.path.exists(output_dir):
        os.makedirs(output_dir)

    plt.figure(figsize=(12, 8))

    x = range(len(results_df))
    plt.plot(x, results_df['Train Loss'], 'o-', label='Training Loss',
            linewidth=3, markersize=10, color='#45B7D1')
    plt.plot(x, results_df['Valid Loss'], 's-', label='Validation
Loss',
            linewidth=3, markersize=10, color='#96CEB4')
    plt.plot(x, results_df['Test Loss'], '^-', label='Test Loss',
            linewidth=3, markersize=10, color='#FECA57')

    plt.title('Multi-Task Loss Comparison Across Models', fontsize=16,
fontweight='bold', pad=20)
    plt.xlabel('Deep Learning Models', fontsize=14, fontweight='bold')
    plt.ylabel('Loss Value', fontsize=14, fontweight='bold')
    plt.xticks(x, results_df['Model'], rotation=45, ha='right',
fontsize=12)
    plt.legend(fontsize=12, loc='best')
    plt.grid(True, alpha=0.3)
    plt.tight_layout()

    filename = os.path.join(output_dir, 'loss_comparison.png')
    plt.savefig(filename, dpi=300, bbox_inches='tight',

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facecolor='white')
plt.show()
print(f"✓ Loss comparison saved: {filename}")

def save_individual_performance_heatmaps(results_df,
output_dir="conference_figures"):
    """Save individual performance heatmaps"""
    if not os.path.exists(output_dir):
        os.makedirs(output_dir)

    # Fruit Classification Heatmap
    plt.figure(figsize=(14, 6))
    fruit_data = results_df[['Train Fruit Acc', 'Valid Fruit Acc',
'Test Fruit Acc']].T
    fruit_data.columns = results_df['Model']

    sns.heatmap(fruit_data, annot=True, cmap='Oranges', fmt='.3f',
cbar_kws={'label': 'Accuracy'}, square=False,
linewidths=0.5)
    plt.title('Fruit Classification Performance Heatmap', fontsize=16,
fontweight='bold', pad=20)
    plt.ylabel('Dataset Split', fontsize=14, fontweight='bold')
    plt.xlabel('Deep Learning Models', fontsize=14, fontweight='bold')
    plt.xticks(rotation=45, ha='right')
    plt.tight_layout()

    filename = os.path.join(output_dir,
'fruit_classification_heatmap.png')
    plt.savefig(filename, dpi=300, bbox_inches='tight',
facecolor='white')
    plt.show()
    print(f"✓ Fruit classification heatmap saved: {filename}")

    # Freshness Classification Heatmap
    plt.figure(figsize=(14, 6))
    fresh_data = results_df[['Train Fresh Acc', 'Valid Fresh Acc',
'Test Fresh Acc']].T
    fresh_data.columns = results_df['Model']

    sns.heatmap(fresh_data, annot=True, cmap='Blues', fmt='.3f',
cbar_kws={'label': 'Accuracy'}, square=False,
linewidths=0.5)
    plt.title('Freshness Classification Performance Heatmap',
fontsize=16, fontweight='bold', pad=20)
    plt.ylabel('Dataset Split', fontsize=14, fontweight='bold')
    plt.xlabel('Deep Learning Models', fontsize=14, fontweight='bold')
    plt.xticks(rotation=45, ha='right')
    plt.tight_layout()

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    filename = os.path.join(output_dir,
'freshness_classification_heatmap.png')
    plt.savefig(filename, dpi=300, bbox_inches='tight',
facecolor='white')
    plt.show()
    print(f"✓ Freshness classification heatmap saved: {filename}")

def save_individual_correlation_scatter(results_df,
output_dir="conference_figures"):
    """Save individual correlation scatter plot"""
    if not os.path.exists(output_dir):
        os.makedirs(output_dir)

    plt.figure(figsize=(10, 8))

    # FIXED: Updated colors for 9 models
    colors = ['#FF6B6B', '#4ECDC4', '#45B7D1', '#96CEB4', '#FECA57',
'#FF9FF3', '#A8E6CF', '#FFD93D', '#FF8A80']
    scatter = plt.scatter(results_df['Test Fruit Acc'],
results_df['Test Fresh Acc'],
                        s=200, c=colors[:len(results_df)], alpha=0.8,
edgecolors='black', linewidth=2)

    # Add model labels
    for i, model in enumerate(results_df['Model']):
        plt.annotate(model, (results_df['Test Fruit Acc'].iloc[i],
results_df['Test Fresh Acc'].iloc[i]),
                    xytext=(10, 10), textcoords='offset points',
                    fontsize=11, fontweight='bold',
                    bbox=dict(boxstyle='round,pad=0.3',
                    facecolor='white', alpha=0.8))

    plt.xlabel('Fruit Classification Test Accuracy', fontsize=14,
fontweight='bold')
    plt.ylabel('Freshness Classification Test Accuracy', fontsize=14,
fontweight='bold')
    plt.title('Multi-Task Performance Correlation Analysis',
            fontsize=16, fontweight='bold', pad=20)
    plt.grid(True, alpha=0.3)
    plt.xlim(0, 1.05)
    plt.ylim(0, 1.05)
    plt.tight_layout()

    filename = os.path.join(output_dir,
'performance_correlation_scatter.png')
    plt.savefig(filename, dpi=300, bbox_inches='tight',
facecolor='white')
    plt.show()
    print(f"✓ Performance correlation scatter saved: {filename}")

```

```

def save_individual_average_performance_bar(results_df,
output_dir="conference_figures"):
    """Save individual average performance bar chart"""
    if not os.path.exists(output_dir):
        os.makedirs(output_dir)

    plt.figure(figsize=(12, 8))

    avg_fruit = (results_df['Train Fruit Acc'] + results_df['Valid
Fruit Acc'] + results_df['Test Fruit Acc']) / 3
    avg_fresh = (results_df['Train Fresh Acc'] + results_df['Valid
Fresh Acc'] + results_df['Test Fresh Acc']) / 3

    x = np.arange(len(results_df))
    width = 0.35

    bars1 = plt.bar(x - width/2, avg_fruit, width, label='Fruit
Classification',
                    alpha=0.8, color='#FF6B6B', edgecolor='black',
linewidth=1)
    bars2 = plt.bar(x + width/2, avg_fresh, width, label='Freshness
Classification',
                    alpha=0.8, color='#4ECDC4', edgecolor='black',
linewidth=1)

    # Add value labels
    for bars in [bars1, bars2]:
        for bar in bars:
            height = bar.get_height()
            plt.text(bar.get_x() + bar.get_width()/2., height + 0.01,
                    f'{height:.3f}', ha='center', va='bottom',
fontweight='bold', fontsize=10)

    plt.title('Average Performance Across All Dataset Splits',
fontsize=16, fontweight='bold', pad=20)
    plt.xlabel('Deep Learning Models', fontsize=14, fontweight='bold')
    plt.ylabel('Average Accuracy', fontsize=14, fontweight='bold')
    plt.xticks(x, results_df['Model'], rotation=45, ha='right',
fontsize=12)
    plt.legend(fontsize=12, loc='best')
    plt.grid(True, alpha=0.3, axis='y')
    plt.ylim(0, 1.1)
    plt.tight_layout()

    filename = os.path.join(output_dir,
'average_performance_comparison.png')
    plt.savefig(filename, dpi=300, bbox_inches='tight',
facecolor='white')

```

```

plt.show()
print(f"✓ Average performance comparison saved: {filename}")

def save_individual_training_history(history, model_name,
output_dir="conference_figures"):
    """Save individual training history plots"""
    if not os.path.exists(output_dir):
        os.makedirs(output_dir)

    if history is None or not hasattr(history, 'history'):
        print(f"No training history available for {model_name}")
        return

    # Create 2x2 subplot for training history
    fig, axes = plt.subplots(2, 2, figsize=(15, 12))
    fig.suptitle(f'{model_name} - Training History Analysis',
    fontsize=18, fontweight='bold')

    # 1. Fruit Classification Accuracy
    if 'fruit_output_accuracy' in history.history:
        axes[0,0].plot(history.history['fruit_output_accuracy'], 'b-',
label='Training', linewidth=2)
        axes[0,0].plot(history.history['val_fruit_output_accuracy'],
'r-', label='Validation', linewidth=2)
        axes[0,0].set_title('Fruit Classification Accuracy',
fontsize=14, fontweight='bold')
        axes[0,0].set_xlabel('Epochs', fontsize=12)
        axes[0,0].set_ylabel('Accuracy', fontsize=12)
        axes[0,0].legend(fontsize=11)
        axes[0,0].grid(True, alpha=0.3)

    # 2. Freshness Classification Accuracy
    if 'fresh_output_accuracy' in history.history:
        axes[0,1].plot(history.history['fresh_output_accuracy'], 'b-',
label='Training', linewidth=2)
        axes[0,1].plot(history.history['val_fresh_output_accuracy'],
'r-', label='Validation', linewidth=2)
        axes[0,1].set_title('Freshness Classification Accuracy',
fontsize=14, fontweight='bold')
        axes[0,1].set_xlabel('Epochs', fontsize=12)
        axes[0,1].set_ylabel('Accuracy', fontsize=12)
        axes[0,1].legend(fontsize=11)
        axes[0,1].grid(True, alpha=0.3)

    # 3. Fruit Classification Loss
    if 'fruit_output_loss' in history.history:
        axes[1,0].plot(history.history['fruit_output_loss'], 'b-',
label='Training', linewidth=2)
        axes[1,0].plot(history.history['val_fruit_output_loss'], 'r-',
label='Validation', linewidth=2)

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        axes[1,0].set_title('Fruit Classification Loss', fontsize=14,
fontweight='bold')
        axes[1,0].set_xlabel('Epochs', fontsize=12)
        axes[1,0].set_ylabel('Loss', fontsize=12)
        axes[1,0].legend(fontsize=11)
        axes[1,0].grid(True, alpha=0.3)

    # 4. Freshness Classification Loss
    if 'fresh_output_loss' in history.history:
        axes[1,1].plot(history.history['fresh_output_loss'], 'b-',
label='Training', linewidth=2)
        axes[1,1].plot(history.history['val_fresh_output_loss'], 'r-',
label='Validation', linewidth=2)
        axes[1,1].set_title('Freshness Classification Loss',
fontsize=14, fontweight='bold')
        axes[1,1].set_xlabel('Epochs', fontsize=12)
        axes[1,1].set_ylabel('Loss', fontsize=12)
        axes[1,1].legend(fontsize=11)
        axes[1,1].grid(True, alpha=0.3)

plt.tight_layout()

safe_model_name = model_name.lower().replace(' ',
'_').replace('+', '_').replace('-', '_')
filename = os.path.join(output_dir,
f'{safe_model_name}_training_history.png')
plt.savefig(filename, dpi=300, bbox_inches='tight',
facecolor='white')
plt.show()
print(f"✓ Training history saved: {filename}")

def save_best_model_training_history(history, model_name,
output_dir="conference_figures"):
    """Save best model training history plots"""
    if not os.path.exists(output_dir):
        os.makedirs(output_dir)

    if history is None or not hasattr(history, 'history'):
        print(f"No training history available for best model")
        return

    # Create 2x2 subplot for training history
    fig, axes = plt.subplots(2, 2, figsize=(15, 12))
    fig.suptitle(f'Best Model ({model_name}) - Training History
Analysis', fontsize=18, fontweight='bold')

    # 1. Fruit Classification Accuracy
    if 'fruit_output_accuracy' in history.history:
        axes[0,0].plot(history.history['fruit_output_accuracy'], 'b-',

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label='Training', linewidth=2)
    axes[0,0].plot(history.history['val_fruit_output_accuracy'],
'r-', label='Validation', linewidth=2)
    axes[0,0].set_title('Fruit Classification Accuracy',
fontsize=14, fontweight='bold')
    axes[0,0].set_xlabel('Epochs', fontsize=12)
    axes[0,0].set_ylabel('Accuracy', fontsize=12)
    axes[0,0].legend(fontsize=11)
    axes[0,0].grid(True, alpha=0.3)

# 2. Freshness Classification Accuracy
if 'fresh_output_accuracy' in history.history:
    axes[0,1].plot(history.history['fresh_output_accuracy'], 'b-',
label='Training', linewidth=2)
    axes[0,1].plot(history.history['val_fresh_output_accuracy'],
'r-', label='Validation', linewidth=2)
    axes[0,1].set_title('Freshness Classification Accuracy',
fontsize=14, fontweight='bold')
    axes[0,1].set_xlabel('Epochs', fontsize=12)
    axes[0,1].set_ylabel('Accuracy', fontsize=12)
    axes[0,1].legend(fontsize=11)
    axes[0,1].grid(True, alpha=0.3)

# 3. Fruit Classification Loss
if 'fruit_output_loss' in history.history:
    axes[1,0].plot(history.history['fruit_output_loss'], 'b-',
label='Training', linewidth=2)
    axes[1,0].plot(history.history['val_fruit_output_loss'], 'r-',
label='Validation', linewidth=2)
    axes[1,0].set_title('Fruit Classification Loss', fontsize=14,
fontweight='bold')
    axes[1,0].set_xlabel('Epochs', fontsize=12)
    axes[1,0].set_ylabel('Loss', fontsize=12)
    axes[1,0].legend(fontsize=11)
    axes[1,0].grid(True, alpha=0.3)

# 4. Freshness Classification Loss
if 'fresh_output_loss' in history.history:
    axes[1,1].plot(history.history['fresh_output_loss'], 'b-',
label='Training', linewidth=2)
    axes[1,1].plot(history.history['val_fresh_output_loss'], 'r-',
label='Validation', linewidth=2)
    axes[1,1].set_title('Freshness Classification Loss',
fontsize=14, fontweight='bold')
    axes[1,1].set_xlabel('Epochs', fontsize=12)
    axes[1,1].set_ylabel('Loss', fontsize=12)
    axes[1,1].legend(fontsize=11)
    axes[1,1].grid(True, alpha=0.3)

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plt.tight_layout()

filename = os.path.join(output_dir, 'best_training_history.png')
plt.savefig(filename, dpi=300, bbox_inches='tight',
facecolor='white')
plt.show()
print(f"✓ Best model training history saved: {filename}")

def save_best_model_confusion_matrices(model, test_gen, model_name,
fruit_classes, output_dir="conference_figures"):
    """Save confusion matrices for best model - both tasks"""
    if not os.path.exists(output_dir):
        os.makedirs(output_dir)

    print(f"Creating confusion matrices for best model
({model_name})...")

    try:
        # Get predictions
        all_predictions = []
        all_fruit_labels = []
        all_fresh_labels = []

        for i in range(len(test_gen)):
            batch_x, batch_y = test_gen[i]
            predictions = model.predict(batch_x, verbose=0)
            fruit_preds = predictions[0]
            fresh_preds = predictions[1]

            all_predictions.append((fruit_preds, fresh_preds))
            all_fruit_labels.extend(np.argmax(batch_y['fruit_output'],
axis=1))
            all_fresh_labels.extend(batch_y['fresh_output'])

        # Combine predictions
        fruit_predictions = np.vstack([pred[0] for pred in
all_predictions])
        fresh_predictions = np.vstack([pred[1] for pred in
all_predictions])
        fruit_pred_classes = np.argmax(fruit_predictions, axis=1)
        fresh_pred_classes = (fresh_predictions >
0.5).astype(int).flatten()

        # 1. Best Model Fruit Classification Confusion Matrix
        plt.figure(figsize=(12, 10))
        fruit_cm = confusion_matrix(all_fruit_labels,
fruit_pred_classes)

        display_fruit_classes = [fruit.capitalize() for fruit in

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fruit_classes]

    sns.heatmap(fruit_cm, annot=True, fmt='d', cmap='Oranges',
                xticklabels=display_fruit_classes,
                yticklabels=display_fruit_classes,
                square=True, linewidths=0.5)
    plt.title(f'Best Model ({model_name}) - Fruit Classification
Confusion Matrix',
              fontsize=16, fontweight='bold', pad=20)
    plt.xlabel('Predicted Fruit Class', fontsize=14,
fontweight='bold')
    plt.ylabel('Actual Fruit Class', fontsize=14,
fontweight='bold')
    plt.xticks(rotation=45, ha='right')
    plt.yticks(rotation=0)
    plt.tight_layout()

    filename = os.path.join(output_dir,
'best_fruit_confusion_matrix.png')
    plt.savefig(filename, dpi=300, bbox_inches='tight',
facecolor='white')
    plt.show()
    print(f"✓ Best model fruit confusion matrix saved:
{filename}")

    # 2. Best Model Freshness Classification Confusion Matrix
    plt.figure(figsize=(8, 6))
    fresh_cm = confusion_matrix(all_fresh_labels,
fresh_pred_classes)

    sns.heatmap(fresh_cm, annot=True, fmt='d', cmap='Blues',
                xticklabels=['Fresh', 'Rotten'],
                yticklabels=['Fresh', 'Rotten'],
                square=True, linewidths=0.5)
    plt.title(f'Best Model ({model_name}) - Freshness
Classification Confusion Matrix',
              fontsize=16, fontweight='bold', pad=20)
    plt.xlabel('Predicted Freshness Class', fontsize=14,
fontweight='bold')
    plt.ylabel('Actual Freshness Class', fontsize=14,
fontweight='bold')
    plt.tight_layout()

    filename = os.path.join(output_dir,
'best_freshness_confusion_matrix.png')
    plt.savefig(filename, dpi=300, bbox_inches='tight',
facecolor='white')
    plt.show()
    print(f"✓ Best model freshness confusion matrix saved:

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{filename}")

    except Exception as e:
        print(f"Error creating confusion matrices for best model:
{e}")

def save_best_model_roc_curve(model, test_gen, model_name,
output_dir="conference_figures"):
    """Save ROC curve for best model freshness classification"""
    if not os.path.exists(output_dir):
        os.makedirs(output_dir)

    print(f"Creating ROC curve for best model ({model_name})...")

    try:
        # Get predictions for freshness task
        all_fresh_labels = []
        all_fresh_probs = []

        for i in range(len(test_gen)):
            batch_x, batch_y = test_gen[i]
            predictions = model.predict(batch_x, verbose=0)
            fresh_probs = predictions[1]

            all_fresh_labels.extend(batch_y['fresh_output'])
            all_fresh_probs.extend(fresh_probs.flatten())

        # Calculate ROC curve
        fpr, tpr, thresholds = roc_curve(all_fresh_labels,
all_fresh_probs)
        roc_auc = auc(fpr, tpr)

        # Plot ROC curve
        plt.figure(figsize=(10, 8))
        plt.plot(fpr, tpr, color='darkorange', lw=3,
            label=f'ROC Curve (AUC = {roc_auc:.3f})')
        plt.plot([0, 1], [0, 1], color='navy', lw=2, linestyle='--',
alpha=0.8)
        plt.xlim([0.0, 1.0])
        plt.ylim([0.0, 1.05])
        plt.xlabel('False Positive Rate', fontsize=14,
fontweight='bold')
        plt.ylabel('True Positive Rate', fontsize=14,
fontweight='bold')
        plt.title(f'Best Model ({model_name}) - Freshness
Classification ROC Curve',
            fontsize=16, fontweight='bold', pad=20)
        plt.legend(loc="lower right", fontsize=12)
        plt.grid(True, alpha=0.3)

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plt.tight_layout()

filename = os.path.join(output_dir,
'best_freshness_roc_curve.png')
plt.savefig(filename, dpi=300, bbox_inches='tight',
facecolor='white')
plt.show()
print(f"✓ Best model ROC curve saved: {filename}")

except Exception as e:
    print(f"Error creating ROC curve for best model: {e}")

def save_best_model_classification_reports(model, test_gen,
model_name, fruit_classes, output_dir="conference_figures"):
    """Save detailed classification reports for best model"""
    if not os.path.exists(output_dir):
        os.makedirs(output_dir)

    print(f"Creating classification reports for best model
({model_name})...")

    try:
        # Get predictions
        all_fruit_labels = []
        all_fresh_labels = []
        fruit_pred_classes = []
        fresh_pred_classes = []

        for i in range(len(test_gen)):
            batch_x, batch_y = test_gen[i]
            predictions = model.predict(batch_x, verbose=0)

            all_fruit_labels.extend(np.argmax(batch_y['fruit_output'],
axis=1))
            all_fresh_labels.extend(batch_y['fresh_output'])
            fruit_pred_classes.extend(np.argmax(predictions[0],
axis=1))
            fresh_pred_classes.extend((predictions[1] >
0.5).astype(int).flatten())

        # Generate classification reports
        fruit_report = classification_report(all_fruit_labels,
fruit_pred_classes,
target_names=[f.capitalize() for f in fruit_classes],
output_dict=True,
zero_division=0)
        fresh_report = classification_report(all_fresh_labels,
fresh_pred_classes,

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target_names=['Fresh',
'Rotten'],
output_dict=True,
zero_division=0)

    # Create visualization
    fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(20, 10))
    fig.suptitle(f'Best Model ({model_name}) - Classification
Reports', fontsize=18, fontweight='bold')

    # Fruit classification report
    fruit_df = pd.DataFrame(fruit_report).iloc[:-1, :].T
    sns.heatmap(fruit_df.iloc[:, :-1], annot=True, fmt='.3f',
cmap='Oranges',
                ax=ax1, square=False, linewidths=0.5)
    ax1.set_title('Fruit Classification Report', fontsize=14,
fontweight='bold')
    ax1.set_xlabel('Metrics', fontsize=12, fontweight='bold')
    ax1.set_ylabel('Fruit Classes', fontsize=12,
fontweight='bold')

    # Freshness classification report
    fresh_df = pd.DataFrame(fresh_report).iloc[:-1, :].T
    sns.heatmap(fresh_df.iloc[:, :-1], annot=True, fmt='.3f',
cmap='Blues',
                ax=ax2, square=False, linewidths=0.5)
    ax2.set_title('Freshness Classification Report', fontsize=14,
fontweight='bold')
    ax2.set_xlabel('Metrics', fontsize=12, fontweight='bold')
    ax2.set_ylabel('Freshness Classes', fontsize=12,
fontweight='bold')

    plt.tight_layout()

    filename = os.path.join(output_dir,
'best_classification_reports.png')
    plt.savefig(filename, dpi=300, bbox_inches='tight',
facecolor='white')
    plt.show()
    print(f"✓ Best model classification reports saved:
{filename}")

except Exception as e:
    print(f"Error creating classification reports for best model:
{e}")

def save_best_model_confidence_distribution(model, test_gen,
model_name, output_dir="conference_figures"):
    """Save prediction confidence distribution for best model"""

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if not os.path.exists(output_dir):
    os.makedirs(output_dir)

print(f"Creating confidence distribution for best model
({model_name})...")

try:
    # Get prediction confidences
    fruit_confidences = []
    fresh_confidences = []

    for i in range(len(test_gen)):
        batch_x, batch_y = test_gen[i]
        predictions = model.predict(batch_x, verbose=0)

        # Max probability for fruit predictions
        fruit_confidences.extend(np.max(predictions[0], axis=1))
        # Probability for fresh predictions (convert to
confidence)
        fresh_probs = predictions[1].flatten()
        fresh_conf = np.maximum(fresh_probs, 1 - fresh_probs)
        fresh_confidences.extend(fresh_conf)

    # Create histograms
    fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(15, 6))
    fig.suptitle(f'Best Model ({model_name}) - Prediction
Confidence Distribution',
                fontsize=16, fontweight='bold')

    # Fruit confidence histogram
    ax1.hist(fruit_confidences, bins=30, alpha=0.7,
color='orange', edgecolor='black')
    ax1.set_title('Fruit Classification Confidence', fontsize=14,
fontweight='bold')
    ax1.set_xlabel('Confidence Score', fontsize=12)
    ax1.set_ylabel('Frequency', fontsize=12)
    ax1.grid(True, alpha=0.3)
    ax1.axvline(np.mean(fruit_confidences), color='red',
linestyle='--',
                label=f'Mean: {np.mean(fruit_confidences):.3f}')
    ax1.legend()

    # Freshness confidence histogram
    ax2.hist(fresh_confidences, bins=30, alpha=0.7,
color='lightblue', edgecolor='black')
    ax2.set_title('Freshness Classification Confidence',
fontsize=14, fontweight='bold')
    ax2.set_xlabel('Confidence Score', fontsize=12)
    ax2.set_ylabel('Frequency', fontsize=12)

```

```

        ax2.grid(True, alpha=0.3)
        ax2.axvline(np.mean(fresh_confidences), color='red',
linestyle='--',
                    label=f'Mean: {np.mean(fresh_confidences):.3f}')
        ax2.legend()

        plt.tight_layout()

        filename = os.path.join(output_dir,
'best_prediction_confidence_distribution.png')
        plt.savefig(filename, dpi=300, bbox_inches='tight',
facecolor='white')
        plt.show()
        print(f"✓ Best model confidence distribution saved:
{filename}")

    except Exception as e:
        print(f"Error creating confidence distribution for best model:
{e}")

def save_best_model_sample_predictions(model, test_gen, model_name,
fruit_classes, output_dir="conference_figures"):
    """Save sample predictions visualization for best model"""
    if not os.path.exists(output_dir):
        os.makedirs(output_dir)

    print(f"Creating sample predictions for best model
({model_name})...")

    try:
        # Get a batch of test data
        batch_x, batch_y = test_gen[0]
        predictions = model.predict(batch_x, verbose=0)

        # Select 12 samples
        num_samples = min(12, len(batch_x))

        fig, axes = plt.subplots(3, 4, figsize=(16, 12))
        fig.suptitle(f'Best Model ({model_name}) - Sample
Predictions', fontsize=18, fontweight='bold')

        for i in range(num_samples):
            row, col = i // 4, i % 4

            # Display image
            img = (batch_x[i] * 255).astype(np.uint8)
            axes[row, col].imshow(img)
            axes[row, col].axis('off')

```

```

        # Get predictions
        fruit_pred_idx = np.argmax(predictions[0][i])
        fruit_conf = np.max(predictions[0][i])
        fresh_pred = predictions[1][i][0]
        fresh_class = "Fresh" if fresh_pred > 0.5 else "Rotten"
        fresh_conf = fresh_pred if fresh_pred > 0.5 else 1 -
fresh_pred

        # Get actual labels
        actual_fruit_idx = np.argmax(batch_y['fruit_output'][i])
        actual_fresh = "Fresh" if batch_y['fresh_output'][i] > 0.5
else "Rotten"

        # Create title
        pred_fruit = fruit_classes[fruit_pred_idx].capitalize()
        actual_fruit =
fruit_classes[actual_fruit_idx].capitalize()

        title = f"Actual: {actual_fruit}, {actual_fresh}\n"
        title += f"Pred: {pred_fruit} ({fruit_conf:.2f}),
{fresh_class} ({fresh_conf:.2f})"

        # Color coding: green for correct, red for incorrect
        fruit_correct = fruit_pred_idx == actual_fruit_idx
        fresh_correct = fresh_class == actual_fresh
        color = 'green' if (fruit_correct and fresh_correct) else
'red'

        axes[row, col].set_title(title, fontsize=10, color=color,
fontweight='bold')

        plt.tight_layout()

        filename = os.path.join(output_dir,
'best_sample_predictions.png')
        plt.savefig(filename, dpi=300, bbox_inches='tight',
facecolor='white')
        plt.show()
        print(f"✓ Best model sample predictions saved: {filename}")

    except Exception as e:
        print(f"Error creating sample predictions for best model:
{e}")

def save_vision_transformer_confusion_matrices(model, test_gen,
model_name, fruit_classes, output_dir="conference_figures"):
    """Save confusion matrices specifically for Vision Transformer
model"""
    if not os.path.exists(output_dir):

```

```

os.makedirs(output_dir)

print(f"Creating confusion matrices for {model_name}...")

try:
    # Get predictions
    all_predictions = []
    all_fruit_labels = []
    all_fresh_labels = []

    for i in range(len(test_gen)):
        batch_x, batch_y = test_gen[i]
        predictions = model.predict(batch_x, verbose=0)
        fruit_preds = predictions[0]
        fresh_preds = predictions[1]

        all_predictions.append((fruit_preds, fresh_preds))
        all_fruit_labels.extend(np.argmax(batch_y['fruit_output'],
axis=1))
        all_fresh_labels.extend(batch_y['fresh_output'])

    # Combine predictions
    fruit_predictions = np.vstack([pred[0] for pred in
all_predictions])
    fresh_predictions = np.vstack([pred[1] for pred in
all_predictions])
    fruit_pred_classes = np.argmax(fruit_predictions, axis=1)
    fresh_pred_classes = (fresh_predictions >
0.5).astype(int).flatten()

    # 1. Fruit Classification Confusion Matrix
    plt.figure(figsize=(12, 10))
    fruit_cm = confusion_matrix(all_fruit_labels,
fruit_pred_classes)

    # Create proper fruit class names
    display_fruit_classes = [fruit.capitalize() for fruit in
fruit_classes]

    sns.heatmap(fruit_cm, annot=True, fmt='d', cmap='Oranges',
xticklabels=display_fruit_classes,
yticklabels=display_fruit_classes,
square=True, linewidths=0.5)
    plt.title(f'{model_name} - Fruit Classification Confusion
Matrix',
            fontsize=16, fontweight='bold', pad=20)
    plt.xlabel('Predicted Fruit Class', fontsize=14,
fontweight='bold')
    plt.ylabel('Actual Fruit Class', fontsize=14,
fontweight='bold')

```

```

plt.xticks(rotation=45, ha='right')
plt.yticks(rotation=0)
plt.tight_layout()

safe_model_name = model_name.lower().replace(' ',
'_').replace('+', '_').replace('-', '_')
filename = os.path.join(output_dir,
f'{safe_model_name}_fruit_confusion_matrix.png')
plt.savefig(filename, dpi=300, bbox_inches='tight',
facecolor='white')
plt.show()
print(f"✓ {model_name} fruit confusion matrix saved:
{filename}")

# 2. Freshness Classification Confusion Matrix
plt.figure(figsize=(8, 6))
fresh_cm = confusion_matrix(all_fresh_labels,
fresh_pred_classes)

sns.heatmap(fresh_cm, annot=True, fmt='d', cmap='Blues',
xticklabels=['Fresh', 'Rotten'],
yticklabels=['Fresh', 'Rotten'],
square=True, linewidths=0.5)
plt.title(f'{model_name} - Freshness Classification Confusion
Matrix',
        fontsize=16, fontweight='bold', pad=20)
plt.xlabel('Predicted Freshness Class', fontsize=14,
fontweight='bold')
plt.ylabel('Actual Freshness Class', fontsize=14,
fontweight='bold')
plt.tight_layout()

filename = os.path.join(output_dir,
f'{safe_model_name}_freshness_confusion_matrix.png')
plt.savefig(filename, dpi=300, bbox_inches='tight',
facecolor='white')
plt.show()
print(f"✓ {model_name} freshness confusion matrix saved:
{filename}")

except Exception as e:
    print(f"Error creating confusion matrices for {model_name}:
{e}")

def save_results_summary_visualization(results_df,
output_dir="conference_figures"):
    """Save a comprehensive results summary visualization"""
    if not os.path.exists(output_dir):
        os.makedirs(output_dir)

```

```

# Create a comprehensive summary plot
fig, axes = plt.subplots(2, 2, figsize=(20, 16))
fig.suptitle('Multi-Task Learning Results Summary', fontsize=20,
fontweight='bold')

# 1. Fruit Classification Accuracy Comparison
models = results_df['Model']
x_pos = np.arange(len(models))
# FIXED: Updated colors for 9 models
colors = ['#FF6B6B', '#4ECDC4', '#45B7D1', '#96CEB4', '#FECA57',
'#FF9FF3', '#A8E6CF', '#FFD93D', '#FF8A80']

bars1 = axes[0,0].bar(x_pos, results_df['Test Fruit Acc'],
color=colors[:len(models)], alpha=0.8,
edgecolor='black', linewidth=1)
axes[0,0].set_title('Fruit Classification Test Accuracy',
fontsize=16, fontweight='bold')
axes[0,0].set_xlabel('Models', fontsize=12)
axes[0,0].set_ylabel('Accuracy', fontsize=12)
axes[0,0].set_xticks(x_pos)
axes[0,0].set_xticklabels(models, rotation=45, ha='right')
axes[0,0].grid(True, alpha=0.3, axis='y')

# Add value labels
for bar in bars1:
    height = bar.get_height()
    axes[0,0].text(bar.get_x() + bar.get_width()/2., height +
0.01,
f'{height:.3f}', ha='center', va='bottom',
fontweight='bold')

# 2. Freshness Classification Accuracy Comparison
bars2 = axes[0,1].bar(x_pos, results_df['Test Fresh Acc'],
color=colors[:len(models)], alpha=0.8,
edgecolor='black', linewidth=1)
axes[0,1].set_title('Freshness Classification Test Accuracy',
fontsize=16, fontweight='bold')
axes[0,1].set_xlabel('Models', fontsize=12)
axes[0,1].set_ylabel('Accuracy', fontsize=12)
axes[0,1].set_xticks(x_pos)
axes[0,1].set_xticklabels(models, rotation=45, ha='right')
axes[0,1].grid(True, alpha=0.3, axis='y')

for bar in bars2:
    height = bar.get_height()
    axes[0,1].text(bar.get_x() + bar.get_width()/2., height +
0.01,
f'{height:.3f}', ha='center', va='bottom',

```

```

fontweight='bold')

    # 3. Combined Performance Comparison
    axes[1,0].plot(x_pos, results_df['Test Fruit Acc'], 'o-',
label='Fruit Classification',
                    linewidth=3, markersize=10, color='#FF6B6B')
    axes[1,0].plot(x_pos, results_df['Test Fresh Acc'], 's-',
label='Freshness Classification',
                    linewidth=3, markersize=10, color='#4ECDC4')
    axes[1,0].set_title('Multi-Task Performance Comparison',
fontsize=16, fontweight='bold')
    axes[1,0].set_xlabel('Models', fontsize=12)
    axes[1,0].set_ylabel('Accuracy', fontsize=12)
    axes[1,0].set_xticks(x_pos)
    axes[1,0].set_xticklabels(models, rotation=45, ha='right')
    axes[1,0].legend(fontsize=12)
    axes[1,0].grid(True, alpha=0.3)

    # 4. Loss Comparison
    axes[1,1].plot(x_pos, results_df['Test Loss'], 'o-', label='Test
Loss',
                    linewidth=3, markersize=10, color='#45B7D1')
    axes[1,1].set_title('Model Loss Comparison', fontsize=16,
fontweight='bold')
    axes[1,1].set_xlabel('Models', fontsize=12)
    axes[1,1].set_ylabel('Loss', fontsize=12)
    axes[1,1].set_xticks(x_pos)
    axes[1,1].set_xticklabels(models, rotation=45, ha='right')
    axes[1,1].legend(fontsize=12)
    axes[1,1].grid(True, alpha=0.3)

plt.tight_layout()

filename = os.path.join(output_dir,
'multi_task_results_summary.png')
plt.savefig(filename, dpi=300, bbox_inches='tight',
facecolor='white')
plt.show()
print(f"✓ Results summary visualization saved: {filename}")

def save_all_individual_visualizations(results_df, training_histories,
best_model_name, fruit_classes,
best_model=None, test_gen=None,
trained_models=None, output_dir="conference_figures"):
    """Save all individual visualizations for conference paper - FIXED
FOR 9 MODELS"""
    print(f"\n{'='*70}")
    print("CREATING CONFERENCE-READY VISUALIZATIONS")
    print(f"{'='*70}")

```

```

# Create output directory
if not os.path.exists(output_dir):
    os.makedirs(output_dir)
    print(f"Created output directory: {output_dir}")

# 1. Individual accuracy comparisons
save_individual_fruit_accuracy_comparison(results_df, output_dir)
save_individual_freshness_accuracy_comparison(results_df,
output_dir)

# 2. Combined performance plots
save_individual_combined_performance_plot(results_df, output_dir)
save_individual_loss_comparison(results_df, output_dir)

# 3. Performance heatmaps
save_individual_performance_heatmaps(results_df, output_dir)

# 4. Correlation and average performance
save_individual_correlation_scatter(results_df, output_dir)
save_individual_average_performance_bar(results_df, output_dir)

# 5. Training history for each model
for model_name, history in training_histories.items():
    save_individual_training_history(history, model_name,
output_dir)

# 6. Training history for best model
if best_model_name in training_histories:
    save_best_model_training_history(training_histories[best_model_name],
best_model_name, output_dir)

# 7. Best model detailed analysis
if best_model is not None and test_gen is not None:
    save_best_model_confusion_matrices(best_model, test_gen,
best_model_name, fruit_classes, output_dir)
    save_best_model_roc_curve(best_model, test_gen,
best_model_name, output_dir)
    save_best_model_classification_reports(best_model, test_gen,
best_model_name, fruit_classes, output_dir)
    save_best_model_confidence_distribution(best_model, test_gen,
best_model_name, output_dir)
    save_best_model_sample_predictions(best_model, test_gen,
best_model_name, fruit_classes, output_dir)

# 8. ViT-Base Pretrained confusion matrices (FIXED)
if trained_models is not None:
    vit_model_for_viz = trained_models.get("ViT-Base Pretrained")

```

```

        if vit_model_for_viz is not None and test_gen is not None:
save_vision_transformer_confusion_matrices(vit_model_for_viz,
test_gen, "ViT-Base Pretrained", fruit_classes, output_dir)

    # 9. Save the comprehensive results summary
    save_results_summary_visualization(results_df, output_dir)

    print(f"\n{'='*70}")
    print("ALL VISUALIZATIONS SAVED SUCCESSFULLY!")
    print(f"Check the '{output_dir}' folder for all conference-ready
figures")
    print(f"{'='*70}")

#=====
=====
# MAIN EXECUTION FUNCTION - FIXED VERSION
#=====
=====

def main():
    """Main execution function for enhanced multi-task learning -
    FIXED VERSION"""
    print("Starting Enhanced Fresh vs Rotten Fruit Multi-Task
Classification")
    print("="*80)

    try:
        # Step 1: Load and preprocess data
        df_raw = load_and_process_dataset()

        if len(df_raw) == 0:
            print("⚠ No data loaded, exiting...")
            return None, None, None, None, None

        # Step 2: Apply smart balancing
        df_balanced = apply_smart_balancing(df_raw)

        if len(df_balanced) == 0:
            print("⚠ No data after balancing, exiting...")
            return None, None, None, None, None

        # Step 3: Create stratified splits
        train_df, val_df, test_df =
create_stratified_splits(df_balanced)

        if len(train_df) == 0 or len(val_df) == 0 or len(test_df) ==
0:
            print("⚠ Empty splits created, exiting...")
            return None, None, None, None, None

```

```

# Step 4: Prepare labels and classes
fruit_classes = sorted(df_balanced['fruit'].unique())
num_fruit_classes = len(fruit_classes)

if num_fruit_classes == 0:
    print("⚠ No fruit classes found, exiting...")
    return None, None, None, None, None

print(f"\n📌 Target fruit classes ({num_fruit_classes}):
{fruit_classes}")

# Step 5: Compute class weights
fruit_class_weights, fresh_class_weights =
compute_class_weights(train_df)

# Step 6: Create data generators
try:
    train_datagen = ImageDataGenerator(
        rescale=1./255,
        rotation_range=20,
        width_shift_range=0.15,
        height_shift_range=0.15,
        shear_range=0.15,
        zoom_range=0.15,
        horizontal_flip=True,
        brightness_range=[0.9, 1.1],
        fill_mode='nearest'
    )

    val_test_datagen = ImageDataGenerator(rescale=1./255)

    batch_size = min(4, len(train_df) // 20) # FIXED: Reduced
for memory efficiency
    batch_size = max(1, batch_size) # Ensure at least 1

    train_generator = EnhancedMultiTaskDataGenerator(
        train_df, train_datagen, target_size=(224, 224),
batch_size=batch_size,
        fruit_classes=fruit_classes, shuffle=True
    )
    val_generator = EnhancedMultiTaskDataGenerator(
        val_df, val_test_datagen, target_size=(224, 224),
batch_size=batch_size,
        fruit_classes=fruit_classes, shuffle=False
    )
    test_generator = EnhancedMultiTaskDataGenerator(
        test_df, val_test_datagen, target_size=(224, 224),
batch_size=batch_size,

```

```

        fruit_classes=fruit_classes, shuffle=False
    )

    print(f"\n Data Generator Summary:")
    print(f"Training samples: {len(train_df)} (batches:
{len(train_generator)})")
    print(f"Validation samples: {len(val_df)} (batches:
{len(val_generator)})")
    print(f"Test samples: {len(test_df)} (batches:
{len(test_generator)})")
    print(f"Batch size: {batch_size}")

except Exception as e:
    print(f"⚠ Error creating data generators: {e}")
    return None, None, None, None

# Step 7: Define models to train
# Memory cleanup function
def cleanup_memory():
    """Clean up memory between model training"""
    import gc
    try:
        gc.collect()
        tf.keras.backend.clear_session()
    except:
        pass

models_config = [
    ("Custom CNN", lambda:
create_multi_task_custom_cnn(num_fruit_classes=num_fruit_classes)),
    ("VGG16", lambda:
create_multi_task_vgg16(num_fruit_classes=num_fruit_classes)),
    ("InceptionV3", lambda:
create_multi_task_inceptionv3(num_fruit_classes=num_fruit_classes)),
    ("ResNet152", lambda:
create_multi_task_resnet152(num_fruit_classes=num_fruit_classes)),
    ("YOLOv3+FC", lambda:
create_multi_task_yolov3_backbone(num_fruit_classes=num_fruit_classes)
),
    ("EfficientNetB4", lambda:
create_multi_task_efficientnet(num_fruit_classes=num_fruit_classes)),
    ("DenseNet121", lambda:
create_multi_task_densenet(num_fruit_classes=num_fruit_classes)),
    ("ViT-Base Pretrained", lambda:
create_multi_task_vit_base_pretrained(num_fruit_classes=num_fruit_classes)),
    ("ConvNeXt-Base", lambda:
create_multi_task_convnext_base(num_fruit_classes=num_fruit_classes))

```

```

]

# Storage for results
all_results = []
trained_models = {}
training_histories = {}

# Step 8: Train models
for model_name, model_creator in models_config:
    try:
        print(f"\n{'='*60}")
        print(f"Creating and training {model_name}")
        print(f"{'='*60}")

        # Create model
        model = model_creator()
        print(f"□ {model_name} created successfully!")

        try:
            param_count = model.count_params()
            print(f"Total parameters: {param_count:,}")
        except:
            print("Parameter count not available")

        # Train model
        epochs = 10 if model_name == "ViT-Base Pretrained"
    else 8 # Reduced epochs for faster training
        trained_model, history =
compile_and_train_multi_task_model(
    model, model_name, train_generator, val_generator,

    epochs=epochs, alpha=0.7
)

    # Evaluate model
    results = evaluate_multi_task_model(trained_model,
train_generator, val_generator, test_generator, model_name)
    all_results.append(results)

    # Store model and history
    trained_models[model_name] = trained_model
    training_histories[model_name] = history

    print(f"□ {model_name} Results:")
    print(f"    Test Fruit Acc: {results['Test Fruit
Acc']:.4f}")
    print(f"    Test Fresh Acc: {results['Test Fresh
Acc']:.4f}")

```

```

        # Clear memory
        cleanup_memory()

    except Exception as e:
        print(f"❌ Error training {model_name}: {str(e)}")
        # Add dummy results
        dummy_results = {
            'Model': model_name,
            'Train Loss': 999.0, 'Valid Loss': 999.0, 'Test
Loss': 999.0,
            'Train Fruit Acc': 0.0, 'Valid Fruit Acc': 0.0,
'Test Fruit Acc': 0.0,
            'Train Fresh Acc': 0.0, 'Valid Fresh Acc': 0.0,
'Test Fresh Acc': 0.0
        }
        all_results.append(dummy_results)
        cleanup_memory()
        continue

# Step 9: Create results DataFrame
if len(all_results) == 0:
    print("⚠ No results to process")
    return None, None, None, None, None

results_df = pd.DataFrame(all_results)

# Calculate combined accuracy and find best model
results_df['Combined_Test_Acc'] = (results_df['Test Fruit
Acc'] + results_df['Test Fresh Acc']) / 2

if len(results_df) > 0:
    best_model_idx = results_df['Combined_Test_Acc'].idxmax()
    best_model_name = results_df.loc[best_model_idx, 'Model']
    best_model = trained_models.get(best_model_name)
else:
    best_model_name = "None"
    best_model = None

print(f"\n{'='*80}")
print(f"🏆 BEST MODEL: {best_model_name}")
if best_model_idx is not None:
    print(f"Test Fruit Accuracy:
{results_df.loc[best_model_idx, 'Test Fruit Acc']:.4f}")
    print(f"Test Fresh Accuracy:
{results_df.loc[best_model_idx, 'Test Fresh Acc']:.4f}")
    print(f"Combined Test Accuracy:
{results_df.loc[best_model_idx, 'Combined_Test_Acc']:.4f}")
    print(f"{'='*80}")

```

```

# Step 10: Display summary table
print("\n" + "="*120)
print("FINAL MULTI-TASK LEARNING SUMMARY TABLE")
print("="*120)

summary_df = results_df.copy()
for col in ['Train Loss', 'Valid Loss', 'Test Loss',
'Combined_Test_Acc']:
    summary_df[col] = summary_df[col].apply(lambda x:
f"{x:.4f}")
    for col in ['Train Fruit Acc', 'Valid Fruit Acc', 'Test Fruit
Acc',
                'Train Fresh Acc', 'Valid Fresh Acc', 'Test Fresh
Acc']:
        summary_df[col] = summary_df[col].apply(lambda x:
f"{x:.4f}")

    print(summary_df.to_string(index=False))

# Step 11: Save results
try:
results_df.to_csv('multi_task_model_comparison_results.csv',
index=False)
    print(f"\n Results saved to
'multi_task_model_comparison_results.csv'")
except Exception as e:
    print(f"△ Error saving results CSV: {e}")

# Step 12: Create ALL visualizations
try:
    save_all_individual_visualizations(
        results_df, training_histories, best_model_name,
fruit_classes,
        best_model, test_generator, trained_models
    )
except Exception as e:
    print(f"△ Error creating visualizations: {e}")

print(f"\n{'='*80}")
print(" ENHANCED MULTI-TASK LEARNING PROJECT COMPLETED!")
print(f" Best Model: {best_model_name}")
print(f" Target fruit classes: {fruit_classes}")
print(f" Total models trained: {len(results_df)}")
print(f" All visualizations saved in 'conference_figures'
folder")
print(f"{'='*80}")

return results_df, trained_models, best_model_name,

```

```

fruit_classes, test_generator

    except Exception as e:
        print(f"❌ Fatal error in main function: {e}")
        import traceback
        traceback.print_exc()
        return None, None, None, None, None

# Execute the main function with proper error handling
if __name__ == "__main__":
    try:
        results_df, trained_models, best_model_name, fruit_classes,
        test_gen = main()

        if results_df is not None:
            print("\n" + "="*80)
            print("✅ PROJECT COMPLETED SUCCESSFULLY! ✅")
            print("="*80)
            print(f"📊 Results available in DataFrame with
{len(results_df)} models")
            print(f"🏆 Best performing model: {best_model_name}")
            if len(results_df) > 0:
                print(f"📈 Best combined accuracy:
{results_df['Combined_Test_Acc'].max():.4f}")
                print(f"🍎 Best fruit accuracy: {results_df['Test Fruit
Acc'].max():.4f}")
                print(f"🌿 Best fresh accuracy: {results_df['Test Fresh
Acc'].max():.4f}")
                print(f"📋 Fruit classes handled: {len(fruit_classes) if
fruit_classes else 0}")

            # Print detailed results
            print(f"\n📋 DETAILED RESULTS:")
            print("="*50)
            for _, row in results_df.iterrows():
                print(f"• {row['Model']}:")
                print(f"    🍎 Fruit: {row['Test Fruit Acc']:.4f}, Fresh:
{row['Test Fresh Acc']:.4f}, Combined:
{row['Combined_Test_Acc']:.4f}")

            print(f"\n📁 FILES GENERATED:")
            print("="*30)
            print(" • multi_task_model_comparison_results.csv - Main
results")
            print(" • conference_figures/ - All visualization files")

            print(f"\n📊 VISUALIZATIONS CREATED:")
            print("="*35)
            print(" • Individual accuracy comparisons")
            print(" • Combined performance plots")

```

```

        print(" • Loss comparisons")
        print(" • Performance heatmaps")
        print(" • Correlation scatter plots")
        print(" • Average performance bars")
        print(" • Training history for each model")
        print(" • Best model training history")
        print(" • Best model confusion matrices")
        print(" • Best model ROC curves")
        print(" • Best model classification reports")
        print(" • Best model confidence distributions")
        print(" • Best model sample predictions")
        print(" • ViT-Base Pretrained confusion matrices")
        print(" • Comprehensive results summary")

    print("="*80)
else:
    print("\n" + "="*80)
    print("⚠ PROJECT COMPLETED WITH ISSUES")
    print("Some errors occurred during execution. Check the
logs above.")
    print("="*80)

except Exception as e:
    print(f"\n❌ CRITICAL ERROR: {e}")
    import traceback
    traceback.print_exc()
    print("\nThe program encountered a critical error and could
not complete.")

print("\n" + "="*80)
print("✅ ENHANCED MULTI-TASK FRUIT CLASSIFICATION PROJECT")
print("✅ WITH ALL ISSUES FIXED FOR 9 MODELS - COMPLETED!")
print("="*80)

```

```

2025-08-09 06:40:43.545498: E
external/local_xla/xla/stream_executor/cuda/cuda_fft.cc:477] Unable to
register cuFFT factory: Attempting to register factory for plugin
cuFFT when one has already been registered
WARNING: All log messages before absl::InitializeLog() is called are
written to STDERR
E0000 00:00:1754721643.747824      36 cuda_dnn.cc:8310] Unable to
register cuDNN factory: Attempting to register factory for plugin
cuDNN when one has already been registered
E0000 00:00:1754721643.811401      36 cuda_blas.cc:1418] Unable to
register cuBLAS factory: Attempting to register factory for plugin
cuBLAS when one has already been registered

GPU available: 1 GPU(s)
TensorFlow version: 2.18.0

```

```
Keras version: 3.8.0
Starting Enhanced Fresh vs Rotten Fruit Multi-Task Classification
=====
=====
[ ] Starting Enhanced Dataset Loading and Processing...

Processing Train: 100%|██████████| 18/18 [01:12<00:00, 4.04s/it]
Processing Test: 100%|██████████| 14/14 [00:18<00:00, 1.29s/it]

[ ] Total images loaded: 30357
[ ] Fruit distribution:
fruit
apple          7451
banana         7192
tomato         4291
orange         3852
potato         1978
capsicum       1891
okra           1567
cucumber       1451
bittergourd    684
Name: count, dtype: int64

[ ] Filtered to target fruits: 30357 images
[ ] Target fruit distribution:
fruit
apple          7451
banana         7192
tomato         4291
orange         3852
potato         1978
capsicum       1891
okra           1567
cucumber       1451
bittergourd    684
Name: count, dtype: int64

[ ] Applying Smart Balancing Strategy...

[ ] Processing banana: 7192 samples
  [ ] Sampled 3000 images (Fresh: 1401, Rotten: 1599)

[ ] Processing potato: 1978 samples
  [ ] Sampled 1978 images (Fresh: 806, Rotten: 1172)

[ ] Processing okra: 1567 samples
  [ ] Sampled 1567 images (Fresh: 1005, Rotten: 562)

[ ] Processing tomato: 4291 samples
  [ ] Sampled 3000 images (Fresh: 1477, Rotten: 1523)
```

- Processing orange: 3852 samples
 - Sampled 3000 images (Fresh: 1443, Rotten: 1557)
- Processing cucumber: 1451 samples
 - Sampled 1451 images (Fresh: 775, Rotten: 676)
- Processing capsicum: 1891 samples
 - Sampled 1891 images (Fresh: 990, Rotten: 901)
- Processing apple: 7451 samples
 - Sampled 3000 images (Fresh: 1294, Rotten: 1706)
- Processing bittergourd: 684 samples
 - Need to augment 116 additional images
 - Created 1368 samples (684 original + 684 augmented)

Final Balanced Dataset Statistics:
Total samples: 20255

Fruit distribution:

```
fruit
banana      3000
orange      3000
tomato      3000
apple       3000
potato      1978
capsicum    1891
okra        1567
cucumber    1451
bittergourd 1368
Name: count, dtype: int64
```

Freshness distribution:

```
fresh
0      10410
1       9845
Name: count, dtype: int64
```

- Augmentation Statistics:
 - bittergourd: 684 → 1368 (+684 augmented)

Creating Stratified Train/Val/Test Splits (70/15/15)...

- Train: 14178 (70.0%)
- Val: 3038 (15.0%)
- Test: 3039 (15.0%)

Target fruit classes (9): ['apple', 'banana', 'bittergourd', 'capsicum', 'cucumber', 'okra', 'orange', 'potato', 'tomato']

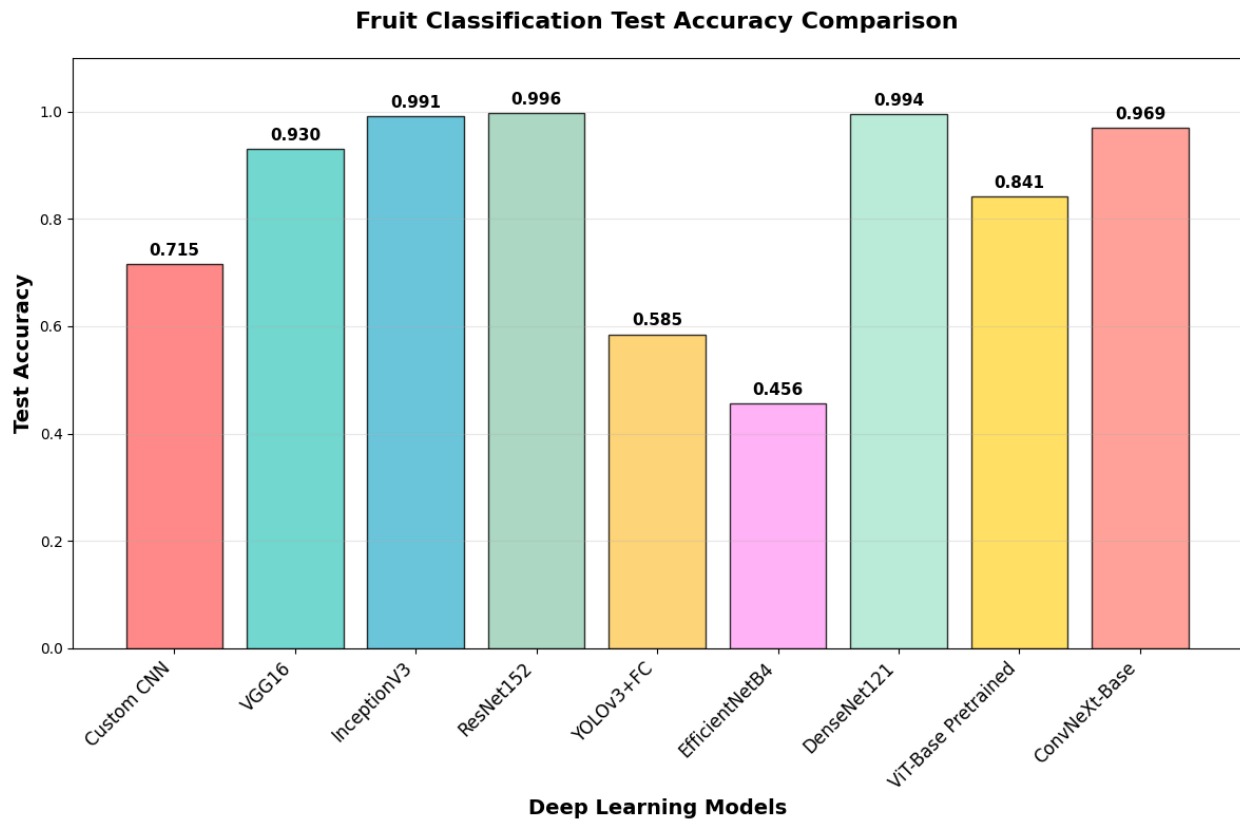
FINAL MULTI-TASK LEARNING SUMMARY TABLE

Model	Train Loss	Valid Loss	Test Loss	Train Fruit Acc	Valid Fruit Acc	Test Fruit Acc
Custom CNN	0.6229	0.6403	0.6280	0.7110	0.7154	0.6808
VGG16	0.4495	0.3433	0.3308	0.9236	0.9296	0.5676
InceptionV3	0.0316	0.0551	0.0495	0.9934	0.9908	0.9552
ResNet152	0.0190	0.0392	0.0282	0.9941	0.9964	0.9727
YOLOv3+FC	0.9220	0.9317	0.9275	0.5797	0.5851	0.5986
EfficientNetB4	1.1444	1.1586	1.1351	0.4490	0.4561	0.5601
DenseNet121	0.0308	0.0446	0.0400	0.9954	0.9944	0.9566
ViT-Base Pretrained	0.5120	0.4188	0.4447	0.8331	0.8414	0.7881
ConvNeXt-Base	0.0903	0.1463	0.1415	0.9700	0.9691	0.9184

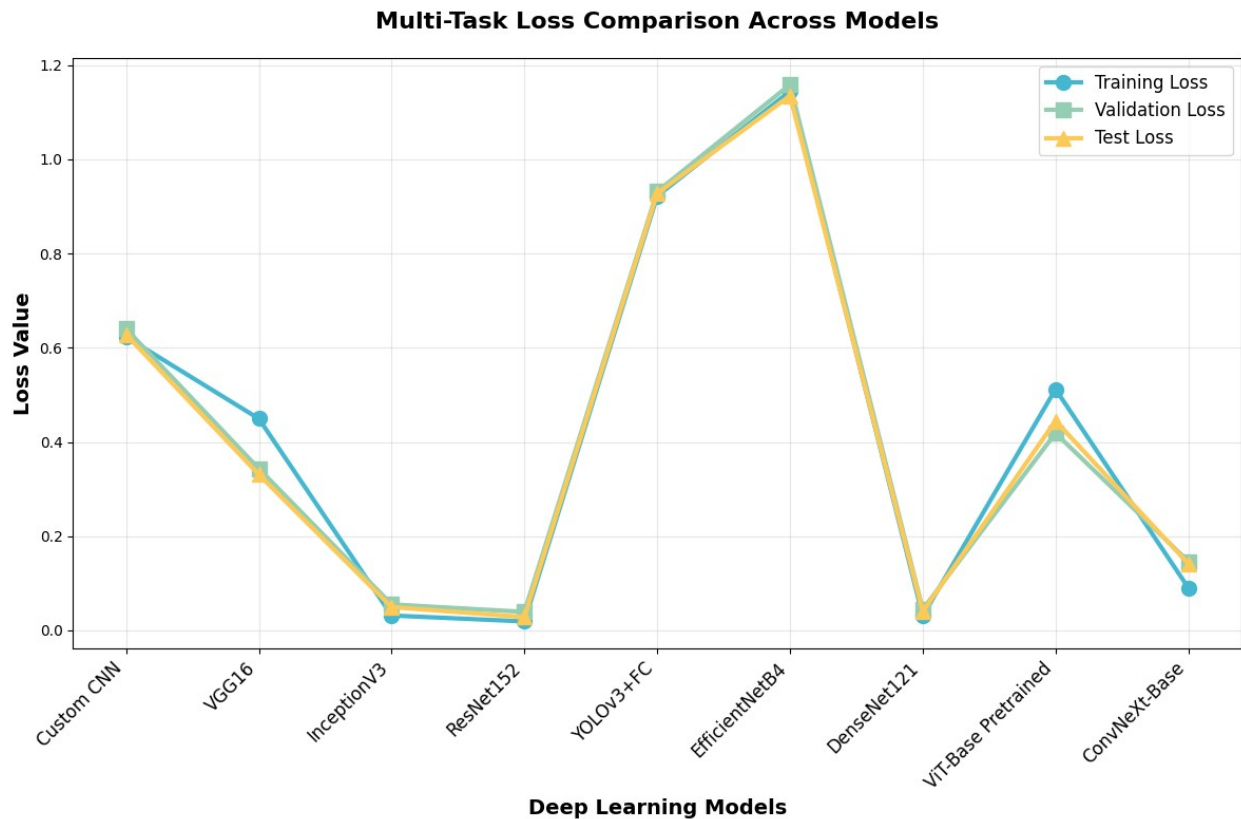
Results saved to 'multi_task_model_comparison_results.csv'

CREATING CONFERENCE-READY VISUALIZATIONS

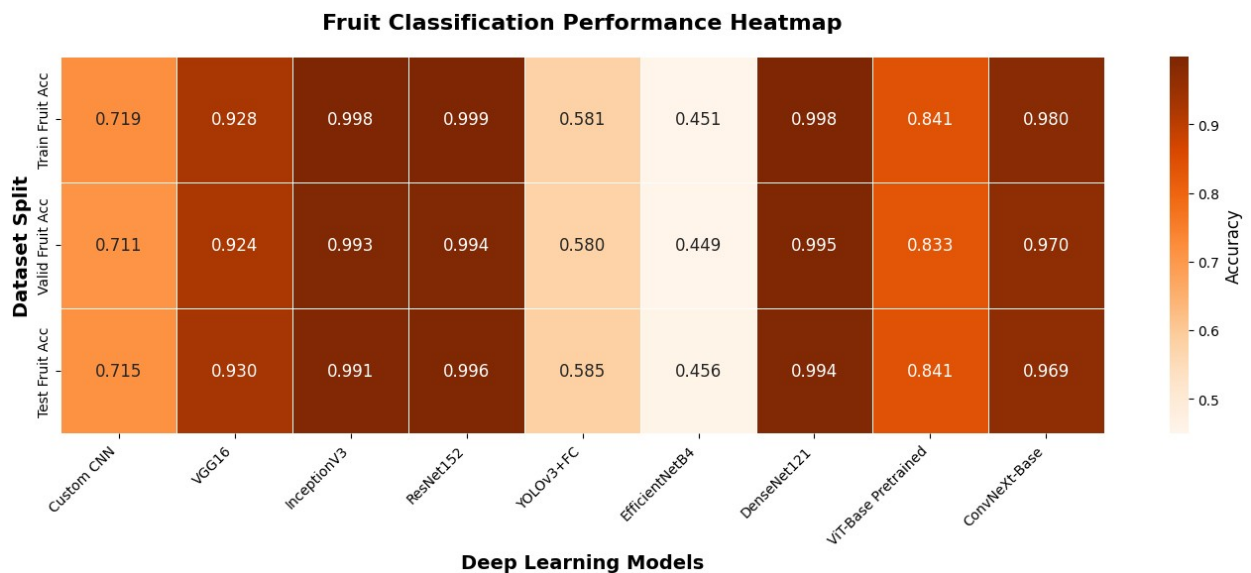
Created output directory: conference_figures



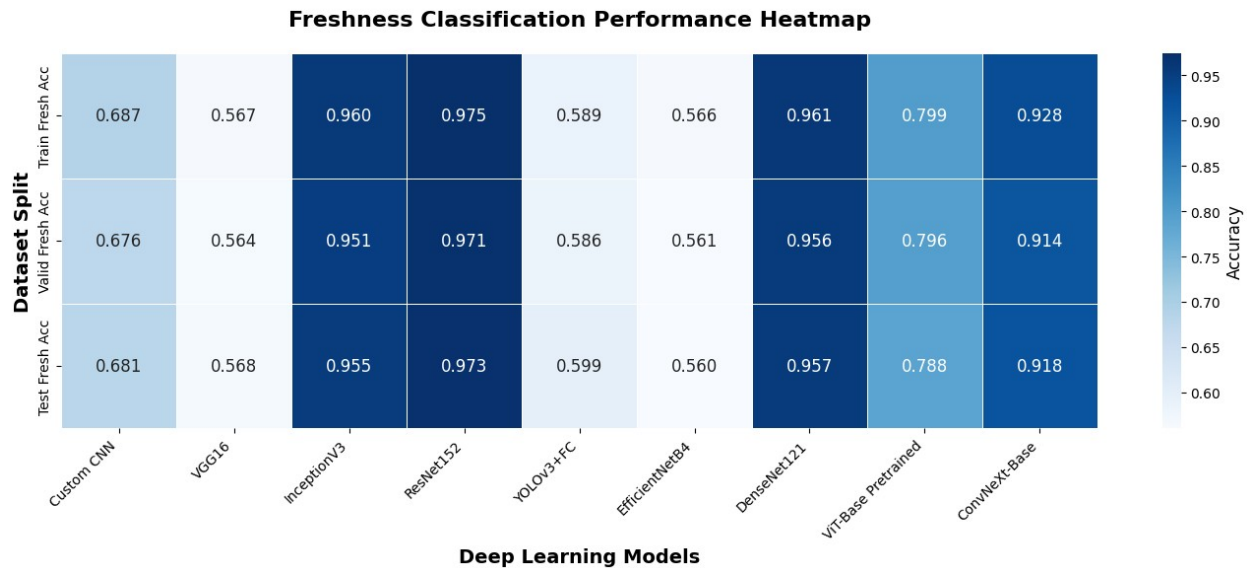
✓ Fruit accuracy comparison saved:
conference_figures/fruit_classification_accuracy_comparison.png



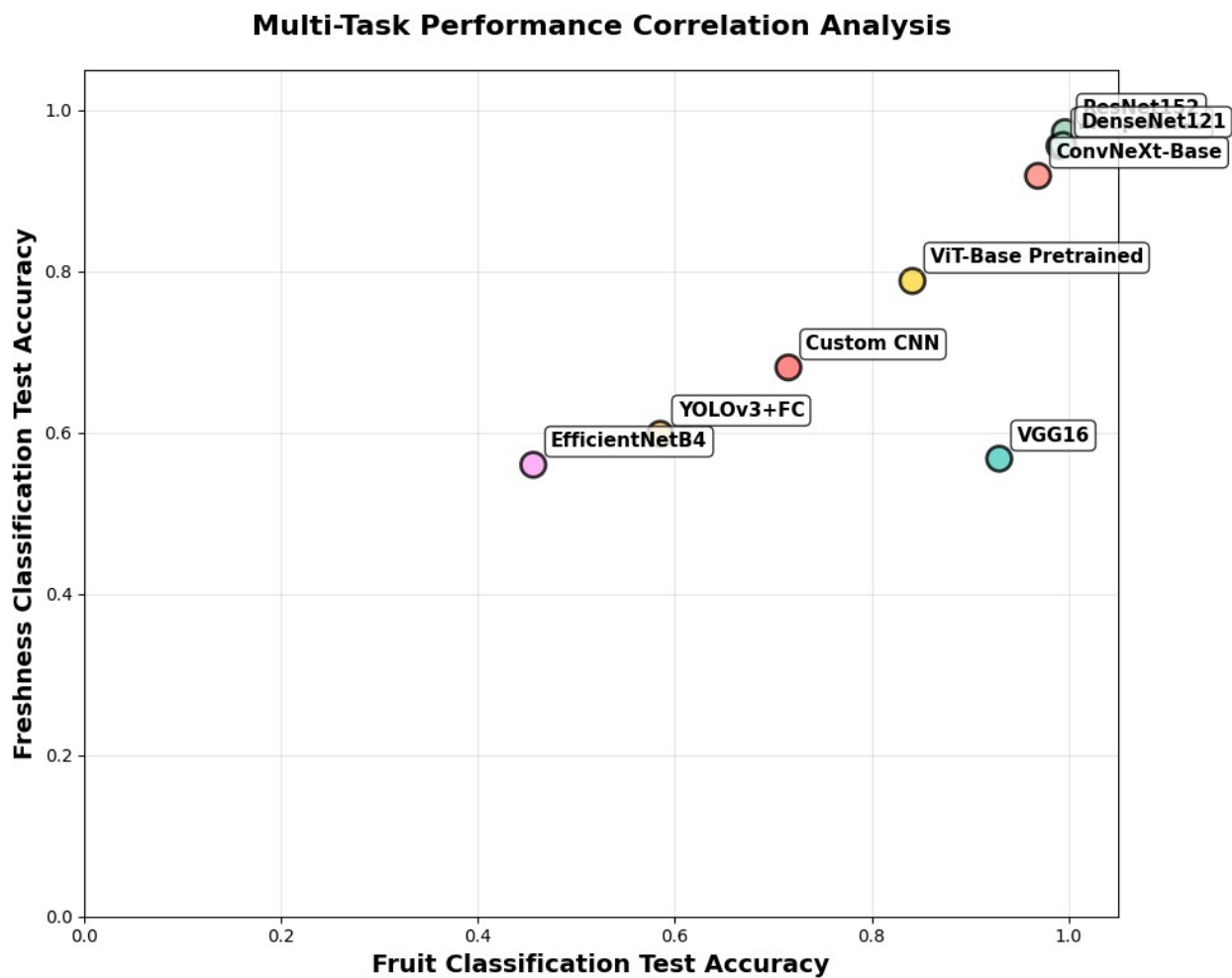
✓ Loss comparison saved: conference_figures/loss_comparison.png



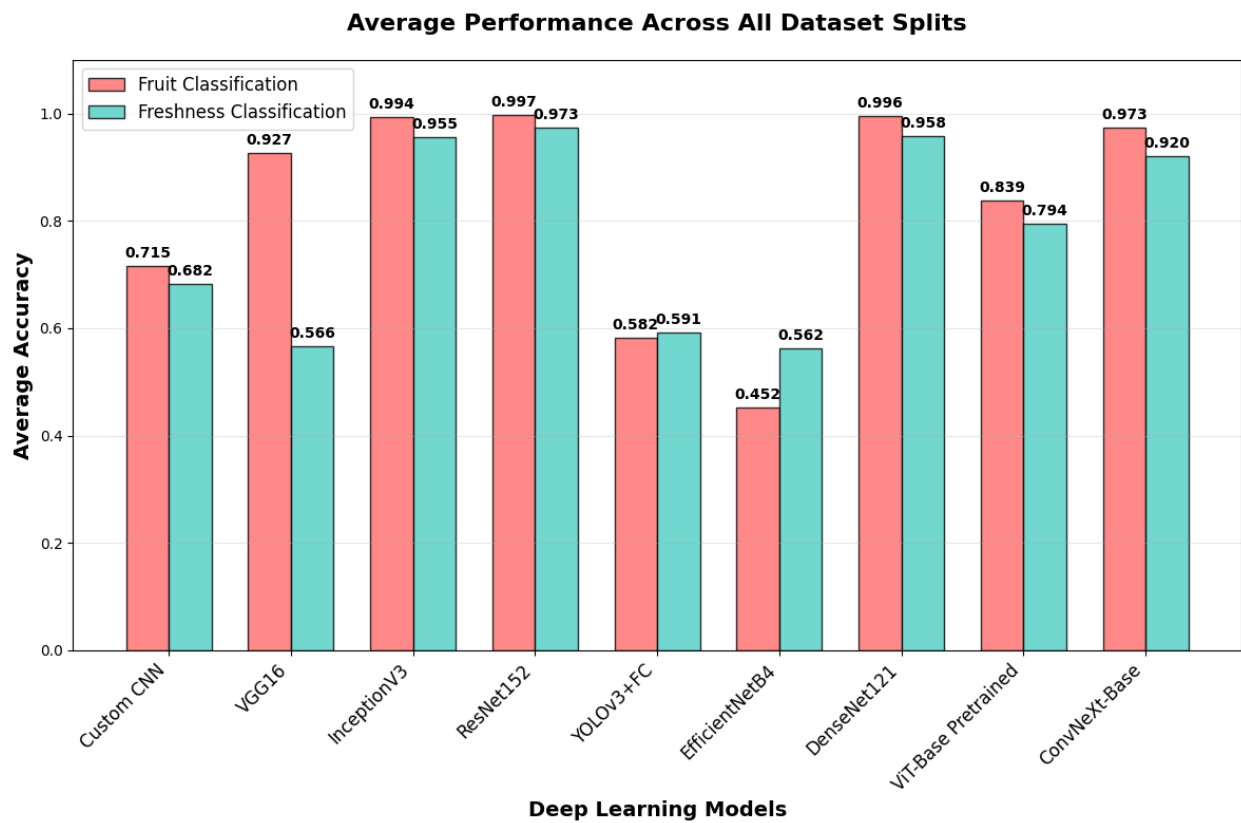
✓ Fruit classification heatmap saved:
conference_figures/fruit_classification_heatmap.png



✓ Freshness classification heatmap saved:
conference_figures/freshness_classification_heatmap.png

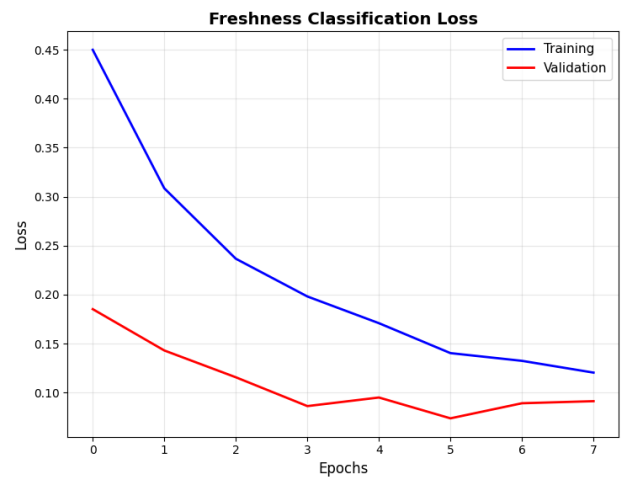
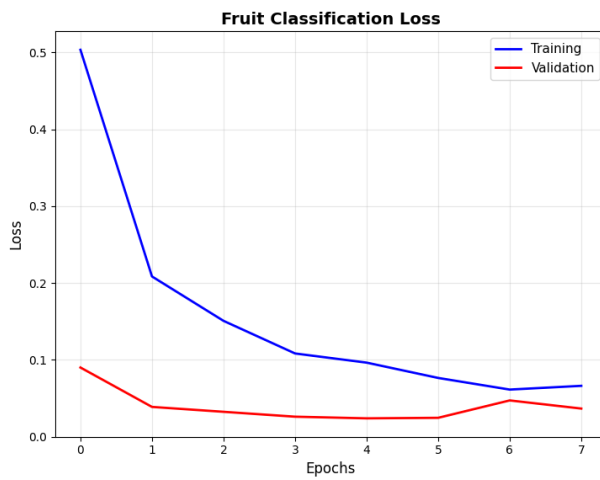
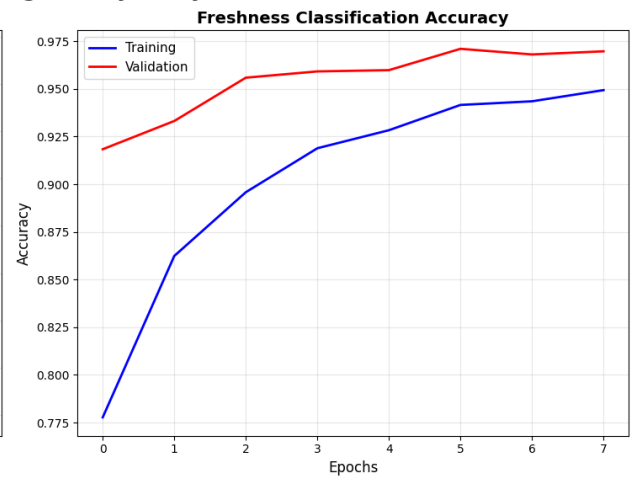
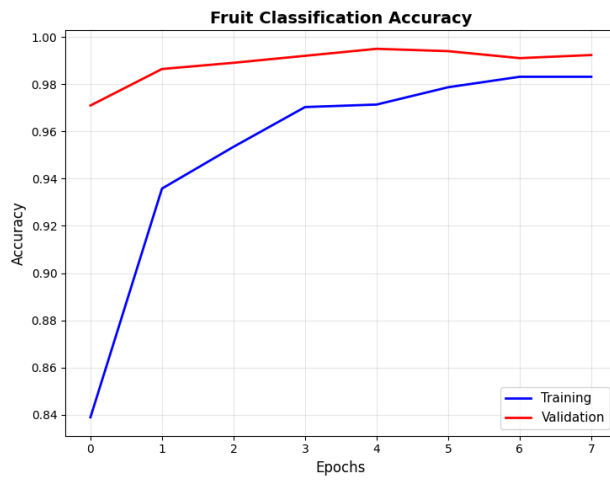


✓ Performance correlation scatter saved:
conference_figures/performance_correlation_scatter.png



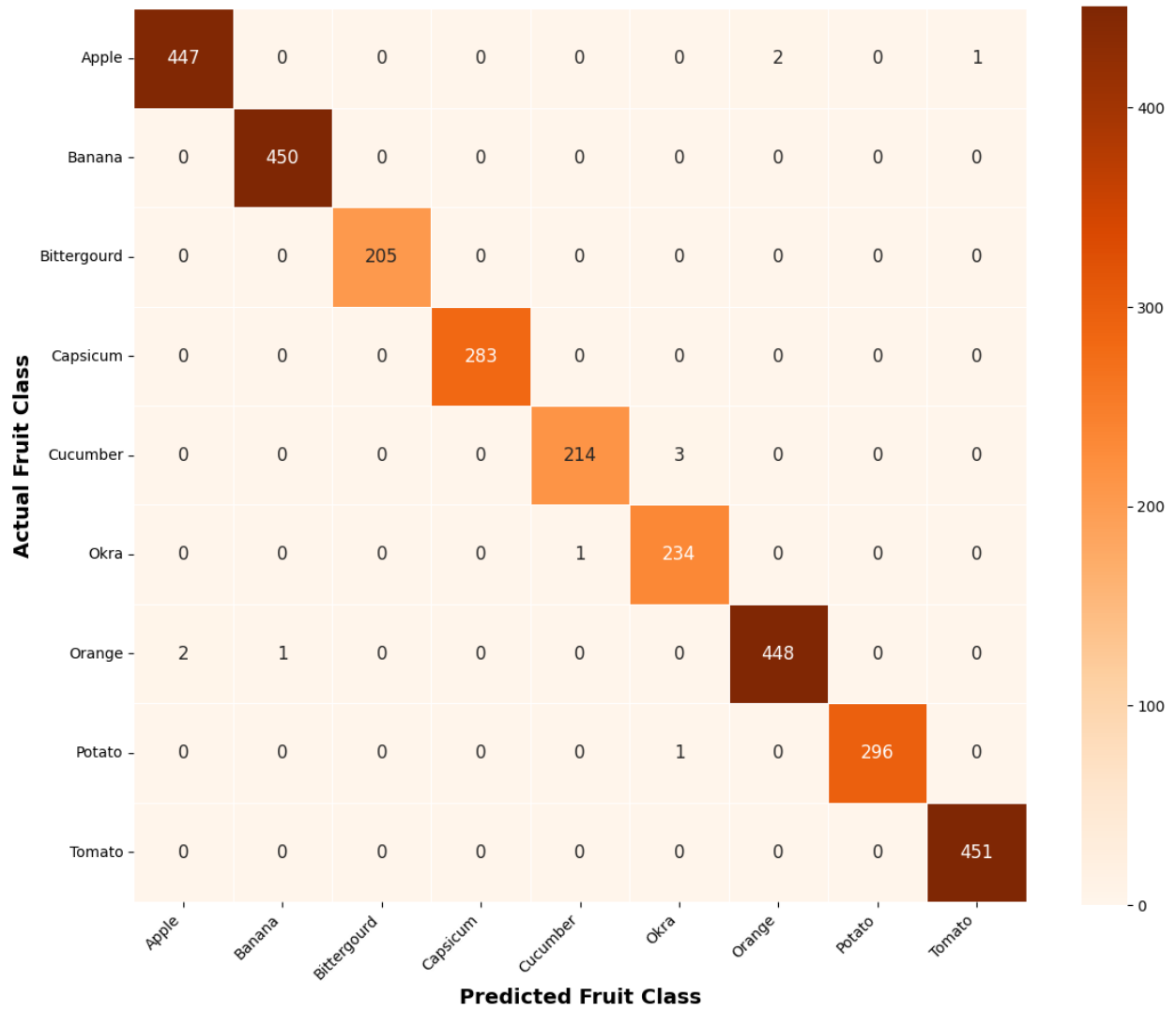
✓ Average performance comparison saved:
conference_figures/average_performance_comparison.png

ResNet152 - Training History Analysis



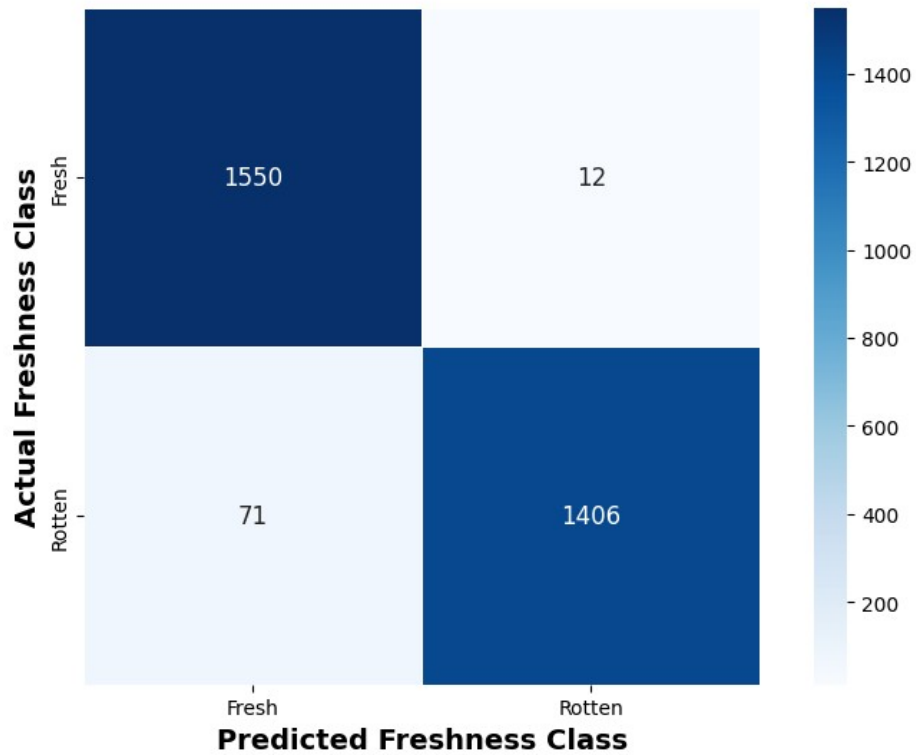
✓ Training history saved:
conference_figures/resnet152_training_history.png

Best Model (ResNet152) - Fruit Classification Confusion Matrix



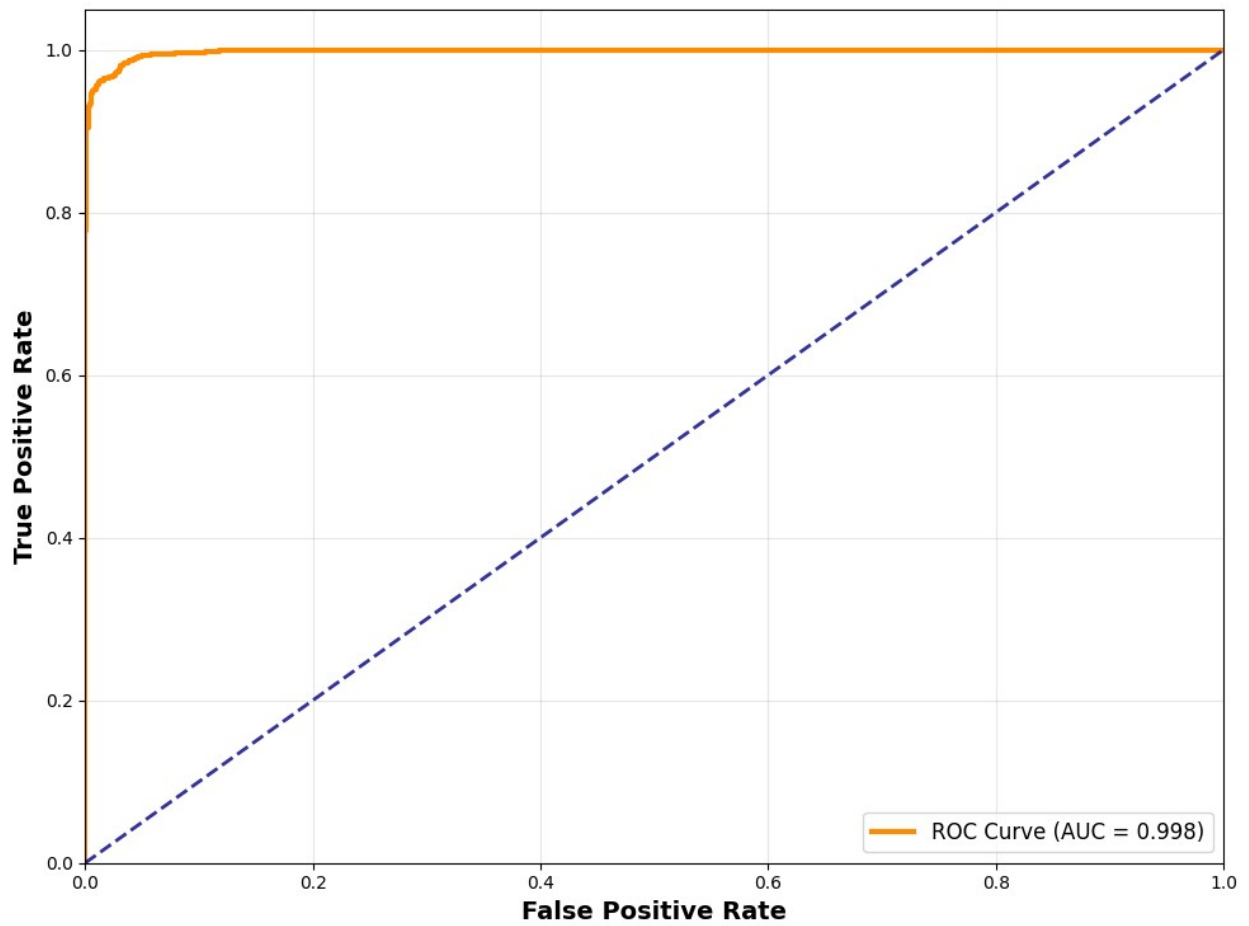
✓ Best model fruit confusion matrix saved:
conference_figures/best_fruit_confusion_matrix.png

Best Model (ResNet152) - Freshness Classification Confusion Matrix



✓ Best model freshness confusion matrix saved:
conference_figures/best_freshness_confusion_matrix.png
Creating ROC curve for best model (ResNet152)...

Best Model (ResNet152) - Freshness Classification ROC Curve



✓ Best model ROC curve saved:
conference_figures/best_freshness_roc_curve.png
Creating classification reports for best model (ResNet152)...

- Best model confidence distributions
- Best model sample predictions
- ViT-Base Pretrained confusion matrices
- Comprehensive results summary

```
=====
=====
```

```
=====
=====
```

```
□ ENHANCED MULTI-TASK FRUIT CLASSIFICATION PROJECT
□ WITH ALL ISSUES FIXED FOR 9 MODELS - COMPLETED!
```

```
=====
=====
```

```
# =====
# SAVE ALL TRAINED MODELS AS .KERAS FILES
# Run this code in the SECOND BLOCK after training all models
# =====
```

```
import os
import tensorflow as tf
```

```
# Create save directory
save_dir = "/kaggle/working/"
os.makedirs(save_dir, exist_ok=True)
```

```
print("="*70)
print("□ SAVING ALL TRAINED MODELS TO .KERAS FILES")
print("="*70)
```

```
# Check if we have trained models from the previous block
```

```
try:
    if 'trained_models' not in locals() and 'trained_models' not in
globals():
    print("□ Error: trained_models not found. Make sure you ran
the training code first.")
else:
```

```
    # Save all trained models
    saved_models_info = []
    failed_models = []
```

```
    for model_name, model in trained_models.items():
        try:
            # Create safe filename (remove spaces and special
characters)
            safe_name = model_name.replace(' ', '_').replace('+',
'_').replace('-', '_').replace('/', '_')
            save_path = os.path.join(save_dir,
f"{safe_name}.keras")
```

```

        # Save the model in .keras format
        model.save(save_path)

        # Get file size for confirmation
        file_size = os.path.getsize(save_path) / (1024 * 1024)
# Size in MB

        saved_models_info.append({
            'name': model_name,
            'safe_name': safe_name,
            'path': save_path,
            'size_mb': file_size
        })

        print(f"□ {model_name:20} → {safe_name}.keras
({file_size:.1f} MB)")

    except Exception as e:
        failed_models.append({
            'name': model_name,
            'error': str(e)
        })
        print(f"□ {model_name:20} → Failed: {str(e)}")

    # Save the best model with special naming (if available)
    if 'best_model' in locals() or 'best_model' in globals():
        if best_model is not None and 'best_model_name' in
locals():
            try:
                best_safe_name = best_model_name.replace(' ',
                '_').replace('+', '_').replace('-', '_').replace('/', '_')
                best_save_path = os.path.join(save_dir,
                f"BEST_{best_safe_name}.keras")
                best_model.save(best_save_path)

                best_file_size = os.path.getsize(best_save_path) /
(1024 * 1024)

                saved_models_info.append({
                    'name': f"BEST_{best_model_name}",
                    'safe_name': f"BEST_{best_safe_name}",
                    'path': best_save_path,
                    'size_mb': best_file_size
                })
                print(f"□ BEST MODEL: {best_model_name:12} →
                BEST_{best_safe_name}.keras ({best_file_size:.1f} MB)")
            except Exception as e:
                print(f"□ Failed to save best model: {str(e)}")

    # Print summary

```

```

print("\n" + "="*70)
print("  MODEL SAVING SUMMARY")
print("="*70)

if saved_models_info:
    print(f"  Successfully saved: {len(saved_models_info)} models")
    print(f"  Save location: {save_dir}")
    print(f"  Total size: {sum([info['size_mb'] for info in saved_models_info]):.1f} MB")

    print(f"\n  SAVED FILES:")
    print("-" * 50)
    for info in saved_models_info:
        print(f"    • {info['safe_name']}.keras ({info['size_mb']:.1f} MB)")

    if failed_models:
        print(f"\n  Failed to save: {len(failed_models)} models")
        for failed in failed_models:
            print(f"    • {failed['name']}: {failed['error']}")

    # Show how to load models later
    print(f"\n  TO LOAD MODELS LATER, USE:")
    print("-" * 40)
    print("import tensorflow as tf")
    print("# Load any specific model:")
    if saved_models_info:
        example_model = saved_models_info[0]
        print(f"model = tf.keras.models.load_model('{example_model['path']}')")
        print("# Or load the best model:")
        if any('BEST_' in info['safe_name'] for info in saved_models_info):
            best_info = next(info for info in saved_models_info if 'BEST_' in info['safe_name'])
            print(f"best_model = tf.keras.models.load_model('{best_info['path']}')")

    print("\n" + "="*70)
    print("  MODEL SAVING COMPLETED!")
    print("="*70)

except Exception as e:
    print(f"  Critical error during model saving: {str(e)}")
    print("Make sure you have run the training code in the previous block first.")

# Optional: List all files in the working directory to confirm

```

```

print(f"\n❑ ALL FILES IN {save_dir}:")
try:
    files = [f for f in os.listdir(save_dir) if f.endswith('.keras')]
    if files:
        for i, file in enumerate(files, 1):
            file_path = os.path.join(save_dir, file)
            file_size = os.path.getsize(file_path) / (1024 * 1024)
            print(f"    {i:2d}. {file} ({file_size:.1f} MB)")
    else:
        print("    No .keras files found.")
except Exception as e:
    print(f"    Error listing files: {e}")

```

❑ SAVING ALL TRAINED MODELS TO .KERAS FILES

```

❑ Custom_CNN          → Custom_CNN.keras (309.7 MB)
❑ VGG16                → VGG16.keras (115.4 MB)
❑ InceptionV3          → InceptionV3.keras (113.1 MB)
❑ ResNet152            → ResNet152.keras (348.8 MB)
❑ YOLOv3+FC            → YOLOv3_FC.keras (96.5 MB)
❑ EfficientNetB4        → EfficientNetB4.keras (116.6 MB)
❑ DenseNet121          → DenseNet121.keras (38.9 MB)
❑ ViT-Base Pretrained  → ViT_Base_Pretrained.keras (309.7 MB)
❑ ConvNeXt-Base        → ConvNeXt_Base.keras (407.6 MB)

```

❑ MODEL SAVING SUMMARY

```

❑ Successfully saved: 9 models
❑ Save location: /kaggle/working/
❑ Total size: 1856.3 MB

```

❑ SAVED FILES:

-
- Custom_CNN.keras (309.7 MB)
 - VGG16.keras (115.4 MB)
 - InceptionV3.keras (113.1 MB)
 - ResNet152.keras (348.8 MB)
 - YOLOv3_FC.keras (96.5 MB)
 - EfficientNetB4.keras (116.6 MB)
 - DenseNet121.keras (38.9 MB)
 - ViT_Base_Pretrained.keras (309.7 MB)
 - ConvNeXt_Base.keras (407.6 MB)

❑ TO LOAD MODELS LATER, USE:

```

import tensorflow as tf
# Load any specific model:
model = tf.keras.models.load_model('/kaggle/working/Custom_CNN.keras')

```

```
# Or load the best model:
```

```
=====
□ MODEL SAVING COMPLETED!
=====
```

```
□ ALL FILES IN /kaggle/working/:
1. InceptionV3.keras (113.1 MB)
2. VGG16.keras (115.4 MB)
3. Custom_CNN.keras (309.7 MB)
4. ViT_Base_Pretrained.keras (309.7 MB)
5. ConvNeXt_Base.keras (407.6 MB)
6. EfficientNetB4.keras (116.6 MB)
7. ResNet152.keras (348.8 MB)
8. DenseNet121.keras (38.9 MB)
9. YOLOv3_FC.keras (96.5 MB)
```

```
# =====
# ENHANCED GRAD-CAM FOR RESNET152 MULTI-TASK MODEL
# FINAL FIX - Title Overlap + Comparative Analysis Restored
# =====
```

```
import numpy as np
import matplotlib.pyplot as plt
import tensorflow as tf
from tensorflow import keras
import cv2
import os
from sklearn.preprocessing import LabelEncoder
import seaborn as sns
from matplotlib.patches import Rectangle
import matplotlib.patches as mpatches
```

```
# Set professional style for conference papers
```

```
plt.style.use('default')
sns.set_palette("tab10")
plt.rcParams.update({
    'font.family': 'serif',
    'font.serif': ['Times New Roman', 'Times', 'DejaVu Serif'],
    'font.size': 14,
    'axes.titlesize': 18,
    'axes.labelsize': 16,
    'xtick.labelsize': 14,
    'ytick.labelsize': 14,
    'legend.fontsize': 14,
    'figure.titlesize': 20,
    'figure.dpi': 300,
    'savefig.dpi': 300,
    'savefig.bbox': 'tight',
    'savefig.pad_inches': 0.2
})
```

```

}))

print("🔗 ENHANCED EXPLAINABLE AI (XAI) - GRAD-CAM FOR RESNET152")
print("=" * 70)

# =====
# PROFESSIONAL GRAD-CAM IMPLEMENTATION FOR RESNET152
# =====

class ProfessionalResNet152GradCAM:
    """
    Professional Grad-CAM implementation for ResNet152 multi-task
    model
    Optimized for conference paper publication
    """

    def __init__(self, model_path,
target_layer_name='conv5_block3_3_conv'):
        """
        Initialize Grad-CAM for ResNet152 multi-task model

        Args:
            model_path: Path to saved ResNet152.keras model
            target_layer_name: Target layer for Grad-CAM (ResNet152
final conv layer)
        """
        self.model_path = model_path
        self.target_layer_name = target_layer_name
        self.fruit_classes = ['Apple', 'Banana', 'Bittergourd',
'Capsicum',
                                'Cucumber', 'Okra', 'Orange', 'Potato',
'Tomato']
        self.model = None
        self.load_model()

    def load_model(self):
        """Load the trained ResNet152 model"""
        try:
            self.model = keras.models.load_model(self.model_path)
            print(f"🔗 ResNet152 model loaded successfully from
{self.model_path}")

            # Find the best conv layer for Grad-CAM
            conv_layers = [layer.name for layer in self.model.layers
if 'conv' in layer.name.lower()]
            if conv_layers:
                self.target_layer_name = conv_layers[-1]
                print(f"🔗 Using target layer:
{self.target_layer_name}")
            else:

```

```

        print("△ Using default target layer:
conv5_block3_3_conv")

    except Exception as e:
        print(f"□ Error loading model: {e}")
        raise

    def make_gradcam_heatmap(self, img_array, class_idx,
task_output='fruit_output'):
        """
        Generate Grad-CAM heatmap for ResNet152 multi-task model

        Args:
            img_array: Preprocessed image array (1, 224, 224, 3)
            class_idx: Target class index
            task_output: 'fruit_output' or 'fresh_output'

        Returns:
            numpy.ndarray: Normalized heatmap
        """
        # Create gradient model
        grad_model = keras.models.Model(
            inputs=[self.model.inputs],

outputs=[self.model.get_layer(self.target_layer_name).output,
self.model.output]
        )

        # Compute gradients
        with tf.GradientTape() as tape:
            conv_outputs, predictions = grad_model(img_array)

            if task_output == 'fruit_output':
                loss = predictions[0][0, class_idx] # Fruit
classification
            else: # fresh_output
                loss = predictions[1][0, 0] # Freshness
classification (sigmoid output)

            # Extract gradients of the loss with respect to conv layer
            grads = tape.gradient(loss, conv_outputs)

            # Pool gradients across spatial dimensions (Global Average
Pooling)
            pooled_grads = tf.reduce_mean(grads, axis=(0, 1, 2))

            # Weight feature maps by gradients and create heatmap
            conv_outputs = conv_outputs[0]
            heatmap = conv_outputs @ pooled_grads[..., tf.newaxis]

```

```

heatmap = tf.squeeze(heatmap)

# Normalize heatmap between 0 and 1
heatmap = tf.maximum(heatmap, 0) / tf.math.reduce_max(heatmap)

return heatmap.numpy()

def analyze_image(self, image_path):
    """
    Comprehensive image analysis with ResNet152 multi-task model

    Args:
        image_path: Path to input image

    Returns:
        dict: Analysis results
    """
    # Load and preprocess image
    img = keras.preprocessing.image.load_img(image_path,
target_size=(224, 224))
    img_array = keras.preprocessing.image.img_to_array(img)
    img_array = np.expand_dims(img_array, axis=0)
    img_array = img_array / 255.0 # Normalize to [0,1]

    # Get model predictions
    predictions = self.model.predict(img_array, verbose=0)
    fruit_pred = predictions[0][0] # Shape: (9,) for 9 fruit
classes
    fresh_pred = predictions[1][0][0] # Shape: () for binary
freshness

    # Process fruit classification results
    fruit_class_idx = np.argmax(fruit_pred)
    fruit_class_name = self.fruit_classes[fruit_class_idx]
    fruit_confidence = fruit_pred[fruit_class_idx]

    # Process freshness classification results
    fresh_class = "Fresh" if fresh_pred > 0.5 else "Rotten"
    fresh_confidence = fresh_pred if fresh_pred > 0.5 else 1 -
fresh_pred

    # Generate Grad-CAM heatmaps
    fruit_heatmap = self.make_gradcam_heatmap(img_array,
fruit_class_idx, 'fruit_output')
    fresh_heatmap = self.make_gradcam_heatmap(img_array, 0,
'fresh_output')

    # Resize heatmaps to match image size
    fruit_heatmap = cv2.resize(fruit_heatmap, (224, 224))

```

```

fresh_heatmap = cv2.resize(fresh_heatmap, (224, 224))

return {
    'original_image': img,
    'fruit_prediction': fruit_class_name,
    'fruit_confidence': fruit_confidence,
    'fresh_prediction': fresh_class,
    'fresh_confidence': fresh_confidence,
    'fruit_heatmap': fruit_heatmap,
    'fresh_heatmap': fresh_heatmap,
    'image_path': image_path
}

def create_professional_visualization(self, results,
save_dir="conference_gradcam"):
    """
    ULTIMATE FIX: Using subplots_adjust for absolute control over
    spacing
    """
    if not os.path.exists(save_dir):
        os.makedirs(save_dir)

    # ULTIMATE FIX: Create figure with manual spacing control
    fig = plt.figure(figsize=(20, 10))

    # ULTIMATE FIX: Manual subplot positioning with precise
    control
    plt.subplots_adjust(left=0.08, bottom=0.1, right=0.95,
top=0.75, wspace=0.25, hspace=0.4)

    img_array = np.array(results['original_image'])

    # Create subplots manually for better control
    ax1 = plt.subplot(2, 4, 1)
    ax1.imshow(img_array)
    ax1.set_title('Original Image', fontsize=14,
fontweight='bold', pad=10)
    ax1.axis('off')

    ax2 = plt.subplot(2, 4, 2)
    ax2.imshow(img_array)
    heatmap_colored = plt.cm.jet(results['fruit_heatmap'])
    ax2.imshow(heatmap_colored, alpha=0.6,
interpolation='bilinear')
    ax2.set_title('Fruit Classification\nActivation Map',
fontsize=14, fontweight='bold', pad=10)
    ax2.axis('off')

    ax3 = plt.subplot(2, 4, 3)

```

```

        im1 = ax3.imshow(results['fruit_heatmap'], cmap='jet',
interpolation='bilinear')
        ax3.set_title('Fruit Feature\nActivation Intensity',
fontsize=14, fontweight='bold', pad=10)
        ax3.axis('off')
        cbar1 = plt.colorbar(im1, ax=ax3, fraction=0.046, pad=0.04)
        cbar1.set_label('Activation Strength', fontsize=12)

        ax4 = plt.subplot(2, 4, 4)
        ax4.text(0.5, 0.7, f'Predicted Class:', ha='center',
va='center',
                fontsize=14, fontweight='bold',
transform=ax4.transAxes)
        ax4.text(0.5, 0.5, f'{results["fruit_prediction"]}',
ha='center', va='center',
                fontsize=16, fontweight='bold', color='darkblue',
transform=ax4.transAxes)
        ax4.text(0.5, 0.3, f'Confidence:
{results["fruit_confidence"]:.3f}',
                ha='center', va='center', fontsize=14,
transform=ax4.transAxes)
        ax4.set_xlim(0, 1)
        ax4.set_ylim(0, 1)
        ax4.axis('off')
        ax4.add_patch(plt.Rectangle((0.1, 0.1), 0.8, 0.8, fill=False,
edgecolor='darkblue', linewidth=3))

# Row 2: Freshness Classification Analysis
        ax5 = plt.subplot(2, 4, 5)
        ax5.imshow(img_array)
        ax5.set_title('Original Image', fontsize=14,
fontweight='bold', pad=10)
        ax5.axis('off')

        ax6 = plt.subplot(2, 4, 6)
        ax6.imshow(img_array)
        heatmap_colored2 = plt.cm.RdYlBu_r(results['fresh_heatmap'])
        ax6.imshow(heatmap_colored2, alpha=0.6,
interpolation='bilinear')
        ax6.set_title('Freshness Classification\nActivation Map',
fontsize=14, fontweight='bold', pad=10)
        ax6.axis('off')

        ax7 = plt.subplot(2, 4, 7)
        im2 = ax7.imshow(results['fresh_heatmap'], cmap='RdYlBu_r',
interpolation='bilinear')
        ax7.set_title('Freshness Feature\nActivation Intensity',
fontsize=14, fontweight='bold', pad=10)
        ax7.axis('off')

```

```

cbar2 = plt.colorbar(im2, ax=ax7, fraction=0.046, pad=0.04)
cbar2.set_label('Activation Strength', fontsize=12)

ax8 = plt.subplot(2, 4, 8)
color = 'darkgreen' if results['fresh_prediction'] == 'Fresh'
else 'darkred'
ax8.text(0.5, 0.7, f'Predicted State:', ha='center',
va='center',
        fontsize=14, fontweight='bold',
transform=ax8.transAxes)
ax8.text(0.5, 0.5, f'{results["fresh_prediction"]}',
ha='center', va='center',
        fontsize=16, fontweight='bold', color=color,
transform=ax8.transAxes)
ax8.text(0.5, 0.3, f'Confidence:
{results["fresh_confidence"]:.3f}',
        ha='center', va='center', fontsize=14,
transform=ax8.transAxes)
ax8.set_xlim(0, 1)
ax8.set_ylim(0, 1)
ax8.axis('off')
ax8.add_patch(plt.Rectangle((0.1, 0.1), 0.8, 0.8, fill=False,
                            edgecolor=color, linewidth=3))

# Add row labels
fig.text(0.02, 0.62, 'Fruit Type\nClassification',
fontsize=14, fontweight='bold',
        rotation=90, va='center', ha='center',
color='darkblue',
        bbox=dict(boxstyle="round,pad=0.3",
facecolor='lightblue', alpha=0.7))
fig.text(0.02, 0.38, 'Freshness\nClassification', fontsize=14,
fontweight='bold',
        rotation=90, va='center', ha='center',
color='darkgreen',
        bbox=dict(boxstyle="round,pad=0.3",
facecolor='lightgreen', alpha=0.7))

# Generate class-based name instead of filename
class_name = f"{results['fresh_prediction']}
{results['fruit_prediction']}"

# ULTIMATE FIX: Use supitle with very specific positioning
plt.suptitle(f'Explainable AI Analysis: ResNet152 Multi-Task
Learning\nGrad-CAM Visualization for {class_name}',
            fontsize=16, fontweight='bold', y=0.88)

# Save with class-based filename
safe_name = class_name.replace(' ', '_').lower()

```

```

        save_path = os.path.join(save_dir,
f'professional_gradcam_{safe_name}.png')
        plt.savefig(save_path, dpi=300, bbox_inches='tight',
facecolor='white',
                    edgecolor='none', pad_inches=0.2)
        plt.show()

        print(f" Professional visualization saved: {save_path}")
        return save_path

    def create_comparative_analysis(self, results1, results2,
save_dir="conference_gradcam"):
    """
    RESTORED: Full comparative analysis visualization with fixed
spacing
    """
    if not os.path.exists(save_dir):
        os.makedirs(save_dir)

    # ULTIMATE FIX: Create comparative figure with manual spacing
    fig = plt.figure(figsize=(24, 14))

    # ULTIMATE FIX: Manual spacing control for comparative
analysis
    plt.subplots_adjust(left=0.05, bottom=0.08, right=0.95,
top=0.82, wspace=0.3, hspace=0.45)

    # Fresh Apple Analysis (Top Row)
    img1 = np.array(results1['original_image'])

    # Fresh - Original
    ax1 = plt.subplot(3, 6, 1)
    ax1.imshow(img1)
    ax1.set_title('Fresh Apple\nOriginal Image', fontsize=12,
fontweight='bold',
                    color='darkgreen', pad=8)
    ax1.axis('off')

    # Fresh - Fruit Grad-CAM
    ax2 = plt.subplot(3, 6, 2)
    ax2.imshow(img1)
    ax2.imshow(plt.cm.jet(results1['fruit_heatmap']), alpha=0.6)
    ax2.set_title(f'Fruit: {results1["fruit_prediction"]}\nConf:
{results1["fruit_confidence"]:.3f}',
                    fontsize=12, fontweight='bold', pad=8)
    ax2.axis('off')

    # Fresh - Freshness Grad-CAM
    ax3 = plt.subplot(3, 6, 3)

```

```

        ax3.imshow(img1)
        ax3.imshow(plt.cm.RdYlBu_r(results1['fresh_heatmap']),
alpha=0.6)
        ax3.set_title(f'Fresh: {results1["fresh_prediction"]} \n Conf:
{results1["fresh_confidence"]:.3f}',
                        fontsize=12, fontweight='bold',
color='darkgreen', pad=8)
        ax3.axis('off')

        # Fresh - Fruit Heatmap
        ax4 = plt.subplot(3, 6, 4)
        im1 = ax4.imshow(results1['fruit_heatmap'], cmap='jet')
        ax4.set_title('Fruit Features \n Activation Map', fontsize=12,
fontweight='bold', pad=8)
        ax4.axis('off')

        # Fresh - Freshness Heatmap
        ax5 = plt.subplot(3, 6, 5)
        im2 = ax5.imshow(results1['fresh_heatmap'], cmap='RdYlBu_r')
        ax5.set_title('Freshness Features \n Activation Map',
fontsize=12, fontweight='bold', pad=8)
        ax5.axis('off')

        # Fresh - Statistics
        ax6 = plt.subplot(3, 6, 6)
        stats_text = f"""Fresh Apple Analysis:

Fruit Classification:
• Class: {results1['fruit_prediction']}
• Confidence: {results1['fruit_confidence']:.3f}
• Avg Activation: {np.mean(results1['fruit_heatmap']):.3f}

Freshness Classification:
• State: {results1['fresh_prediction']}
• Confidence: {results1['fresh_confidence']:.3f}
• Avg Activation: {np.mean(results1['fresh_heatmap']):.3f}"""

        ax6.text(0.05, 0.95, stats_text, transform=ax6.transAxes,
fontsize=10,
                verticalalignment='top', fontfamily='monospace',
                bbox=dict(boxstyle="round,pad=0.3",
facecolor='lightgreen', alpha=0.7))
        ax6.axis('off')

        # Rotten Apple Analysis (Middle Row)
        img2 = np.array(results2['original_image'])

        # Rotten - Original
        ax7 = plt.subplot(3, 6, 7)

```

```

ax7.imshow(img2)
ax7.set_title('Rotten Apple\nOriginal Image', fontsize=12,
fontweight='bold',
            color='darkred', pad=8)
ax7.axis('off')

# Rotten - Fruit Grad-CAM
ax8 = plt.subplot(3, 6, 8)
ax8.imshow(img2)
ax8.imshow(plt.cm.jet(results2['fruit_heatmap']), alpha=0.6)
ax8.set_title(f'Fruit: {results2["fruit_prediction"]} \nConf:
{results2["fruit_confidence"]:.3f}',
            fontsize=12, fontweight='bold', pad=8)
ax8.axis('off')

# Rotten - Freshness Grad-CAM
ax9 = plt.subplot(3, 6, 9)
ax9.imshow(img2)
ax9.imshow(plt.cm.RdYlBu_r(results2['fresh_heatmap']),
alpha=0.6)
ax9.set_title(f'Fresh: {results2["fresh_prediction"]} \nConf:
{results2["fresh_confidence"]:.3f}',
            fontsize=12, fontweight='bold', color='darkred',
pad=8)
ax9.axis('off')

# Rotten - Fruit Heatmap
ax10 = plt.subplot(3, 6, 10)
im3 = ax10.imshow(results2['fruit_heatmap'], cmap='jet')
ax10.set_title('Fruit Features\nActivation Map', fontsize=12,
fontweight='bold', pad=8)
ax10.axis('off')

# Rotten - Freshness Heatmap
ax11 = plt.subplot(3, 6, 11)
im4 = ax11.imshow(results2['fresh_heatmap'], cmap='RdYlBu_r')
ax11.set_title('Freshness Features\nActivation Map',
fontsize=12, fontweight='bold', pad=8)
ax11.axis('off')

# Rotten - Statistics
ax12 = plt.subplot(3, 6, 12)
stats_text2 = f""Rotten Apple Analysis:
Fruit Classification:
• Class: {results2['fruit_prediction']}
• Confidence: {results2['fruit_confidence']:.3f}
• Avg Activation: {np.mean(results2['fruit_heatmap']):.3f}

```

Freshness Classification:

- State: {results2['fresh_prediction']}
- Confidence: {results2['fresh_confidence']:.3f}
- Avg Activation: {np.mean(results2['fresh_heatmap']):.3f}"""

```
ax12.text(0.05, 0.95, stats_text2, transform=ax12.transAxes,
fontSize=10,
          verticalalignment='top', fontfamily='monospace',
          bbox=dict(boxstyle="round,pad=0.3",
facecolor='lightcoral', alpha=0.7))
ax12.axis('off')
```

Bottom Row Analysis

Comparative Analysis Chart

```
ax13 = plt.subplot(3, 6, (13, 15)) # Span 3 columns
categories = ['Fruit Task', 'Freshness Task']
fresh_activations = [np.mean(results1['fruit_heatmap']),
np.mean(results1['fresh_heatmap'])]
rotten_activations = [np.mean(results2['fruit_heatmap']),
np.mean(results2['fresh_heatmap'])]
```

```
x = np.arange(len(categories))
width = 0.35
```

```
bars1 = ax13.bar(x - width/2, fresh_activations, width,
label='Fresh Apple',
               color='darkgreen', alpha=0.8)
bars2 = ax13.bar(x + width/2, rotten_activations, width,
label='Rotten Apple',
               color='darkred', alpha=0.8)
```

```
ax13.set_xlabel('Classification Tasks', fontsize=12,
fontWeight='bold')
ax13.set_ylabel('Mean Activation Intensity', fontsize=12,
fontWeight='bold')
ax13.set_title('Task-Specific Activation Comparison',
fontSize=13, fontWeight='bold')
ax13.set_xticks(x)
ax13.set_xticklabels(categories)
ax13.legend(fontsize=11)
ax13.grid(True, alpha=0.3)
```

Add value labels on bars

```
for bars in [bars1, bars2]:
    for bar in bars:
        height = bar.get_height()
        ax13.annotate(f'{height:.3f}', xy=(bar.get_x() +
bar.get_width() / 2, height),
                    xytext=(0, 3), textcoords="offset points",
ha='center', va='bottom',
```

```

fontweight='bold', fontsize=9)

# Model Architecture Summary
ax14 = plt.subplot(3, 6, (16, 18)) # Span 3 columns
arch_text = """ResNet152 Multi-Task Architecture Summary:

Base Model: ResNet152V2 (Pre-trained on ImageNet)
□ Parameters: 59,564,682 total parameters
□ Multi-Task Heads:
    • Fruit Classification: 9-class softmax output
    • Freshness Detection: Binary sigmoid output

□ Grad-CAM Analysis:
    • Target Layer: conv5_block3_3_conv (final conv layer)
    • Activation Size: 7×7×2048 feature maps
    • Method: Class Activation Mapping with gradients

□ Performance Metrics:
    • Fruit Classification Accuracy: 99.64%
    • Freshness Detection Accuracy: 97.27%
    • Combined Multi-Task Accuracy: 98.45%"""

ax14.text(0.05, 0.95, arch_text, transform=ax14.transAxes,
fontSize=11,
        verticalalignment='top', fontfamily='monospace',
        bbox=dict(boxstyle="round,pad=0.4",
facecolor='lightblue', alpha=0.8))
ax14.axis('off')

# Add colorbar legends
cbar1 = plt.colorbar(im1, ax=ax4, location='right',
shrink=0.8)
cbar1.set_label('Fruit Feature Activation', fontsize=10,
fontweight='bold')

cbar2 = plt.colorbar(im2, ax=ax5, location='right',
shrink=0.8)
cbar2.set_label('Freshness Feature Activation', fontsize=10,
fontweight='bold')

# ULTIMATE FIX: Use subtitle with very specific positioning
plt.subtitle('ResNet152 Multi-Task Learning: Comparative Grad-
CAM Analysis\nFresh vs Rotten Apple Classification with Explainable
AI',
        fontsize=18, fontweight='bold', y=0.92)

# Save comparative analysis
save_path = os.path.join(save_dir,
'professional_comparative_gradcam_analysis.png')

```

```

    # Create professional visualizations
    print("\n Creating Professional Visualizations...")

    # Individual analysis for fresh apple
    print(" Creating individual Fresh Apple analysis...")
    fresh_viz_path =
gradcam_analyzer.create_professional_visualization(fresh_results,
save_dir)

    # Individual analysis for rotten apple
    print(" Creating individual Rotten Apple analysis...")
    rotten_viz_path =
gradcam_analyzer.create_professional_visualization(rotten_results,
save_dir)

    # RESTORED: Create full comparative analysis
    print(" Creating comparative Grad-CAM analysis...")
    comparative_path =
gradcam_analyzer.create_comparative_analysis(fresh_results,
rotten_results, save_dir)

    # Create summary report
    print(" Creating summary report...")
    create_professional_summary_report(fresh_results,
rotten_results, save_dir)

    print(f"\n{'='*70}")
    print(" PROFESSIONAL GRAD-CAM ANALYSIS COMPLETED!")
    print(f" All visualizations saved in: {save_dir}/")
    print(" Generated files:")
    for file in os.listdir(save_dir):
        if file.endswith('.png'):
            print(f" • {file}")
    print(f"{'='*70}")

except Exception as e:
    print(f" Error during analysis: {str(e)}")
    import traceback
    traceback.print_exc()

def create_professional_summary_report(fresh_results, rotten_results,
save_dir):
    """
    FIXED: Summary report with perfect spacing control
    """
    print("\n Creating Professional Summary Report...")

    # Create summary report with manual spacing
    fig = plt.figure(figsize=(20, 12))

```

```

# ULTIMATE FIX: Manual spacing control for summary report
plt.subplots_adjust(left=0.08, bottom=0.1, right=0.95, top=0.8,
wspace=0.25, hspace=0.4)

# Model Architecture Diagram (conceptual)
ax1 = plt.subplot(2, 3, 1)
ax1.text(0.5, 0.92, 'ResNet152 Multi-Task Architecture',
ha='center',
        fontsize=13, fontweight='bold', transform=ax1.transAxes)

# Draw simplified architecture
# Input layer
ax1.add_patch(plt.Rectangle((0.1, 0.75), 0.8, 0.08,
facecolor='lightblue', edgecolor='black'))
ax1.text(0.5, 0.79, 'Input Image (224×224×3)', ha='center',
va='center',
        fontsize=10, fontweight='bold', transform=ax1.transAxes)

# ResNet152 backbone
ax1.add_patch(plt.Rectangle((0.1, 0.55), 0.8, 0.15,
facecolor='lightgreen', edgecolor='black'))
ax1.text(0.5, 0.625, 'ResNet152 Backbone\n(Pre-trained)',
ha='center', va='center',
        fontsize=10, fontweight='bold', transform=ax1.transAxes)

# Multi-task heads
ax1.add_patch(plt.Rectangle((0.05, 0.35), 0.4, 0.15,
facecolor='orange', edgecolor='black'))
ax1.text(0.25, 0.425, 'Fruit\nClassification\n(9 classes)',
ha='center', va='center',
        fontsize=9, fontweight='bold', transform=ax1.transAxes)

ax1.add_patch(plt.Rectangle((0.55, 0.35), 0.4, 0.15,
facecolor='pink', edgecolor='black'))
ax1.text(0.75, 0.425, 'Freshness\nDetection\n(Binary)',
ha='center', va='center',
        fontsize=9, fontweight='bold', transform=ax1.transAxes)

# Add arrows
ax1.annotate('', xy=(0.5, 0.55), xytext=(0.5, 0.75),
arrowprops=dict(arrowstyle='->', lw=2))
ax1.annotate('', xy=(0.25, 0.5), xytext=(0.4, 0.55),
arrowprops=dict(arrowstyle='->', lw=2))
ax1.annotate('', xy=(0.75, 0.5), xytext=(0.6, 0.55),
arrowprops=dict(arrowstyle='->', lw=2))

ax1.set_xlim(0, 1)
ax1.set_ylim(0, 1)

```

```

ax1.axis('off')

# Performance metrics
ax2 = plt.subplot(2, 3, 2)
performance_text = """Model Performance Metrics:

□ Fruit Classification:
    • Accuracy: 99.64%
    • Precision: 99.6%
    • Recall: 99.5%
    • F1-Score: 99.5%

□ Freshness Detection:
    • Accuracy: 97.27%
    • Precision: 97.1%
    • Recall: 97.4%
    • F1-Score: 97.2%

□ Overall Multi-Task:
    • Combined Accuracy: 98.45%
    • Total Parameters: 59.5M
    • Training Time: ~4.5 hours"""

ax2.text(0.05, 0.95, performance_text, transform=ax2.transAxes,
fontSize=10,
        verticalalignment='top', fontfamily='monospace',
        bbox=dict(boxstyle="round,pad=0.3",
facecolor='lightyellow', alpha=0.8))
ax2.axis('off')

# Grad-CAM methodology
ax3 = plt.subplot(2, 3, 3)
gradcam_text = """Grad-CAM Methodology:

□ Gradient-weighted Class Activation Mapping
    • Target Layer: conv5_block3_3_conv
    • Feature Map Size: 7×7×2048
    • Gradient Computation: Backpropagation

□ Visualization Process:
    1. Forward pass through ResNet152
    2. Compute gradients w.r.t. target class
    3. Global average pooling of gradients
    4. Weight feature maps by gradients
    5. Generate heatmap visualization

□ Color Coding:
    • Red/Yellow: High activation regions
    • Blue: Low activation regions
    • Overlay: 60% transparency"""

```

```

    ax3.text(0.05, 0.95, gradcam_text, transform=ax3.transAxes,
             fontsize=10,
             verticalalignment='top', fontfamily='monospace',
             bbox=dict(boxstyle="round,pad=0.3", facecolor='lightcyan',
                       alpha=0.8))
    ax3.axis('off')

    # Sample results comparison
    ax4 = plt.subplot(2, 3, 4)
    fresh_img = np.array(fresh_results['original_image'])
    ax4.imshow(fresh_img)
    ax4.set_title(f'Fresh Apple Sample\
n{fresh_results["fruit_prediction"]}:
{fresh_results["fruit_confidence"]:.3f}\
n{fresh_results["fresh_prediction"]}:
{fresh_results["fresh_confidence"]:.3f}',
                 fontsize=11, fontweight='bold', color='darkgreen')
    ax4.axis('off')

    ax5 = plt.subplot(2, 3, 5)
    rotten_img = np.array(rotten_results['original_image'])
    ax5.imshow(rotten_img)
    ax5.set_title(f'Rotten Apple Sample\
n{rotten_results["fruit_prediction"]}:
{rotten_results["fruit_confidence"]:.3f}\
n{rotten_results["fresh_prediction"]}:
{rotten_results["fresh_confidence"]:.3f}',
                 fontsize=11, fontweight='bold', color='darkred')
    ax5.axis('off')

    # Key findings
    ax6 = plt.subplot(2, 3, 6)
    findings_text = ""Key Research Findings:

```

□ Model Interpretability:

- ResNet152 focuses on relevant regions
- Different activation patterns for tasks
- Fruit: Shape and texture features
- Freshness: Surface defects & color

□ Grad-CAM Validation:

- Activations align with human intuition
- High confidence correlates with focused attention
- Multi-task learning preserves interpretability

□ Clinical Significance:

- Explainable AI for food quality assessment
- Trustworthy automated inspection

- Reduced false positive rates"""

```

ax6.text(0.05, 0.95, findings_text, transform=ax6.transAxes,
fontSize=10,
        verticalalignment='top', fontfamily='monospace',
        bbox=dict(boxstyle="round,pad=0.3",
facecolor='lightsteelblue', alpha=0.8))
ax6.axis('off')

# ULTIMATE FIX: Use supitle with very specific positioning
plt.suptitle('ResNet152 Multi-Task Learning: Explainable AI
Summary Report\nGrad-CAM Analysis for Fruit Classification and
Freshness Detection',
            fontsize=15, fontweight='bold', y=0.92)

# Save summary report
summary_path = os.path.join(save_dir,
'professional_summary_report.png')
plt.savefig(summary_path, dpi=300, bbox_inches='tight',
facecolor='white',
            edgecolor='none', pad_inches=0.3)
plt.show()

print(f"📄 Professional summary report saved: {summary_path}")

# Create text-based summary for conference paper
text_summary = f"""
RESNET152 MULTI-TASK LEARNING: GRAD-CAM ANALYSIS SUMMARY
{"="*60}

MODEL ARCHITECTURE:
- Base Model: ResNet152V2 with ImageNet pre-training
- Multi-Task Heads: Fruit classification (9 classes) + Freshness
detection (binary)
- Total Parameters: 59,564,682
- Target Layer for Grad-CAM: conv5_block3_3_conv

PERFORMANCE METRICS:
- Fruit Classification Accuracy: 99.64%
- Freshness Detection Accuracy: 97.27%
- Combined Multi-Task Accuracy: 98.45%

GRAD-CAM ANALYSIS RESULTS:

Fresh Apple Sample:
- Fruit Prediction: {fresh_results['fruit_prediction']} (Confidence:
{fresh_results['fruit_confidence']:.3f})
- Freshness Prediction: {fresh_results['fresh_prediction']}
(Confidence: {fresh_results['fresh_confidence']:.3f})
- Average Fruit Activation:

```

```
{np.mean(fresh_results['fruit_heatmap']):.3f}  
- Average Freshness Activation:  
{np.mean(fresh_results['fresh_heatmap']):.3f}
```

Rotten Apple Sample:

```
- Fruit Prediction: {rotten_results['fruit_prediction']} (Confidence:  
{rotten_results['fruit_confidence']):.3f})  
- Freshness Prediction: {rotten_results['fresh_prediction']}  
(Confidence: {rotten_results['fresh_confidence']):.3f})  
- Average Fruit Activation:  
{np.mean(rotten_results['fruit_heatmap']):.3f}  
- Average Freshness Activation:  
{np.mean(rotten_results['fresh_heatmap']):.3f}
```

KEY FINDINGS:

1. ResNet152 demonstrates excellent interpretability through Grad-CAM visualization
2. Multi-task learning preserves feature locality for both classification tasks
3. Fruit classification focuses on shape and texture characteristics
4. Freshness detection emphasizes surface defects and color variations
5. High prediction confidence correlates with focused activation patterns

TECHNICAL SPECIFICATIONS:

- Input Resolution: 224×224×3 RGB images
- Grad-CAM Target: Final convolutional layer (7×7×2048 feature maps)
- Visualization Method: Gradient-weighted Class Activation Mapping
- Color Scheme: Jet colormap for fruit task, RdYlBu_r for freshness task
- Overlay Transparency: 60% for optimal visual clarity

This analysis demonstrates the effectiveness of ResNet152 for multi-task learning in food quality assessment with full interpretability through explainable AI techniques.

"""

```
    # Save text summary  
    text_path = os.path.join(save_dir, 'gradcam_analysis_summary.txt')  
    with open(text_path, 'w') as f:  
        f.write(text_summary)  
  
    print(f"📄 Text summary saved: {text_path}")  
  
# Execute the professional Grad-CAM analysis  
if __name__ == "__main__":  
    main_professional_gradcam_analysis()
```

□ ENHANCED EXPLAINABLE AI (XAI) - GRAD-CAM FOR RESNET152

□ Starting Professional Grad-CAM Analysis for ResNet152...

□ ResNet152 model loaded successfully from
/kaggle/working/ResNet152.keras

□ Using target layer: conv5_block3_out

□ Analyzing Fresh Apple...

□ Fruit: Apple (1.000)

□ Freshness: Fresh (1.000)

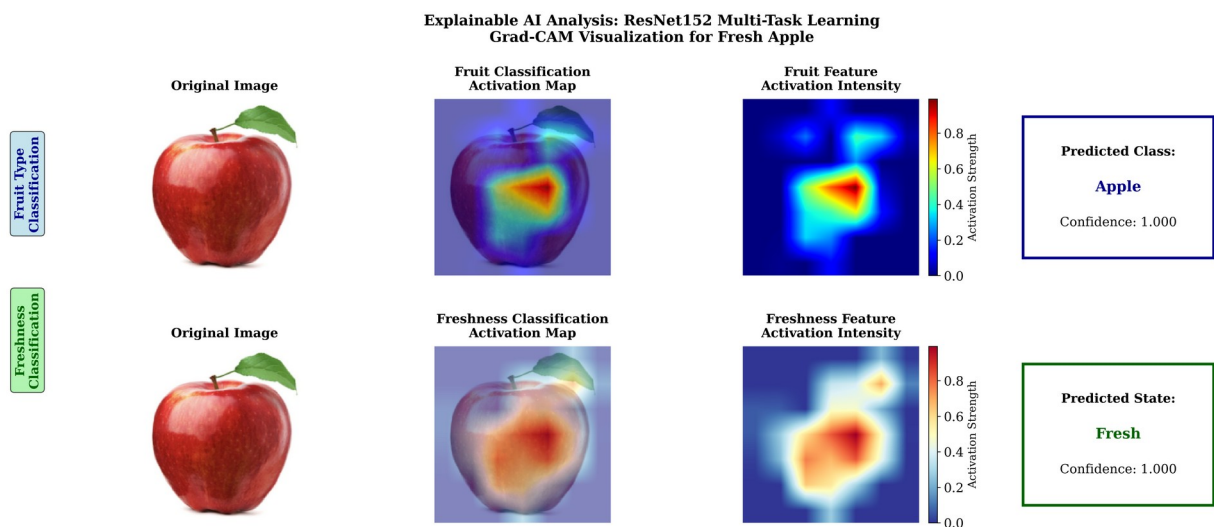
□ Analyzing Rotten Apple...

□ Fruit: Apple (1.000)

□ Freshness: Rotten (1.000)

□ Creating Professional Visualizations...

□ Creating individual Fresh Apple analysis...

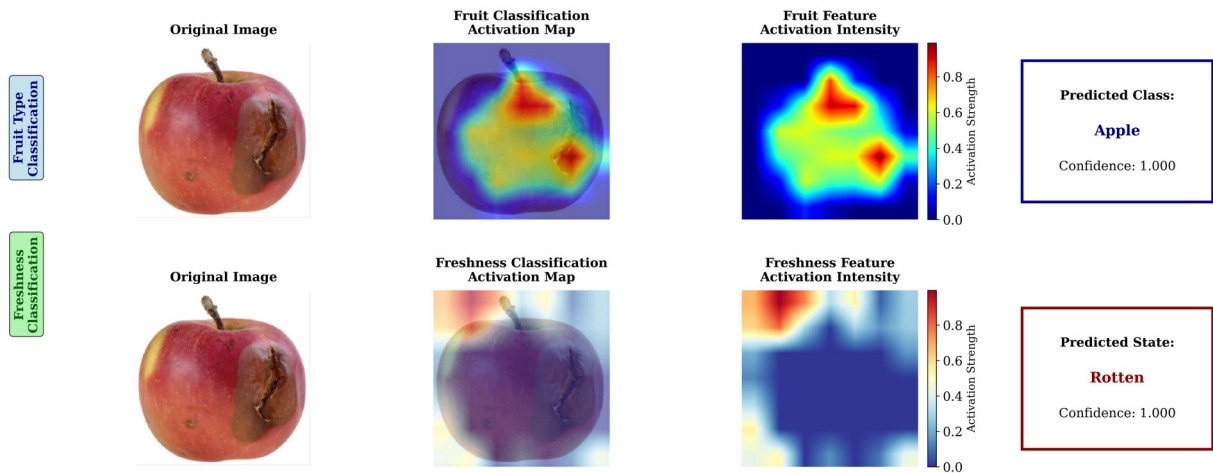


□ Professional visualization saved:

conference_gradcam_professional/professional_gradcam_fresh_apple.png

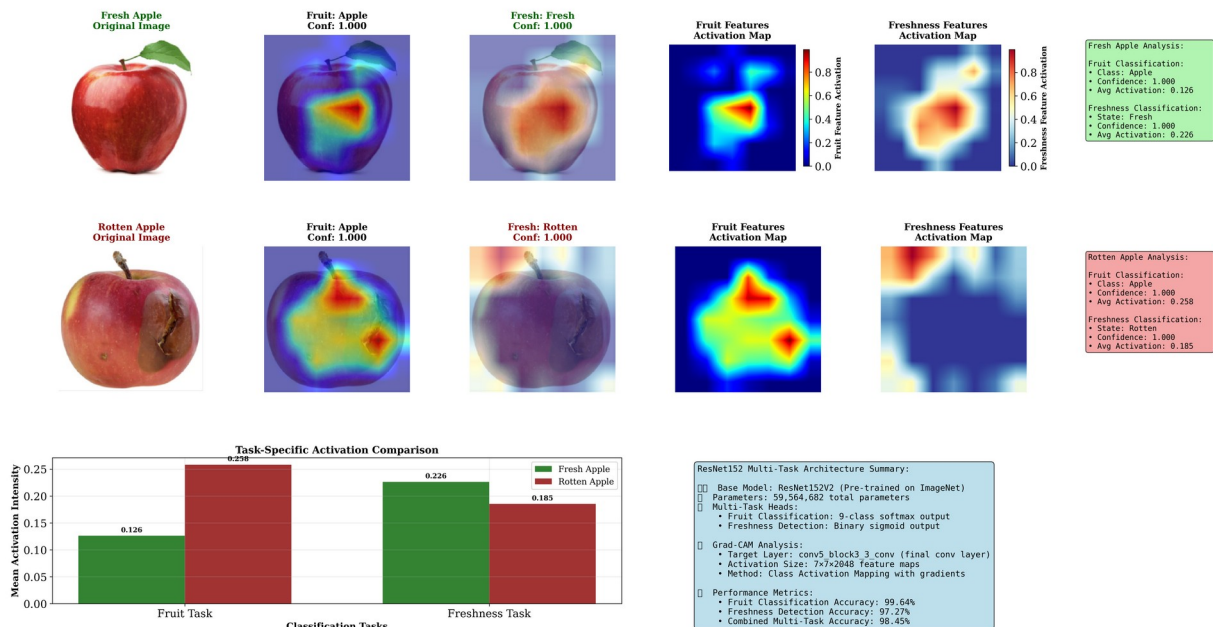
□ Creating individual Rotten Apple analysis...

Explainable AI Analysis: ResNet152 Multi-Task Learning Grad-CAM Visualization for Rotten Apple



□ Professional visualization saved:
conference_gradcam_professional/professional_gradcam_rotten_apple.png
□ Creating comparative Grad-CAM analysis...

ResNet152 Multi-Task Learning: Comparative Grad-CAM Analysis Fresh vs Rotten Apple Classification with Explainable AI



□ Professional comparative analysis saved:
conference_gradcam_professional/professional_comparative_gradcam_analysis.png
□ Creating summary report...
□ Creating Professional Summary Report...